Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF COMPUTER ENGINEERING AND INFORMATION SYSTEM

DEPARTMENT: SOFTWARE ENGINEERING

NAME: VIGNESHWARAN T

REG NO: 23MIS0587

COURSE NAME: Agile Development Process and Devops Lab

COURSE CODE: ISWE406P

SLOT: L3+L4

DIGITAL ASSIGNMENT – II

(**JENKINS**)

JENKINS

1. Learning Goal

Understand **what Jenkins is** and run a job successfully.

Stepwise Tasks

1. What is CI/CD? (Real-world example)

   CI/CD = Continuous Integration / Continuous Delivery

   It's a way for software to **build, test, and deploy automatically** whenever code changes.

   ### Real-World Example

   Imagine a group project where:

   - One person writes code
   - Another tests it
   - Another uploads it to the server

   Without CI/CD, this is manual, slow, and full of "it works on my laptop" drama.

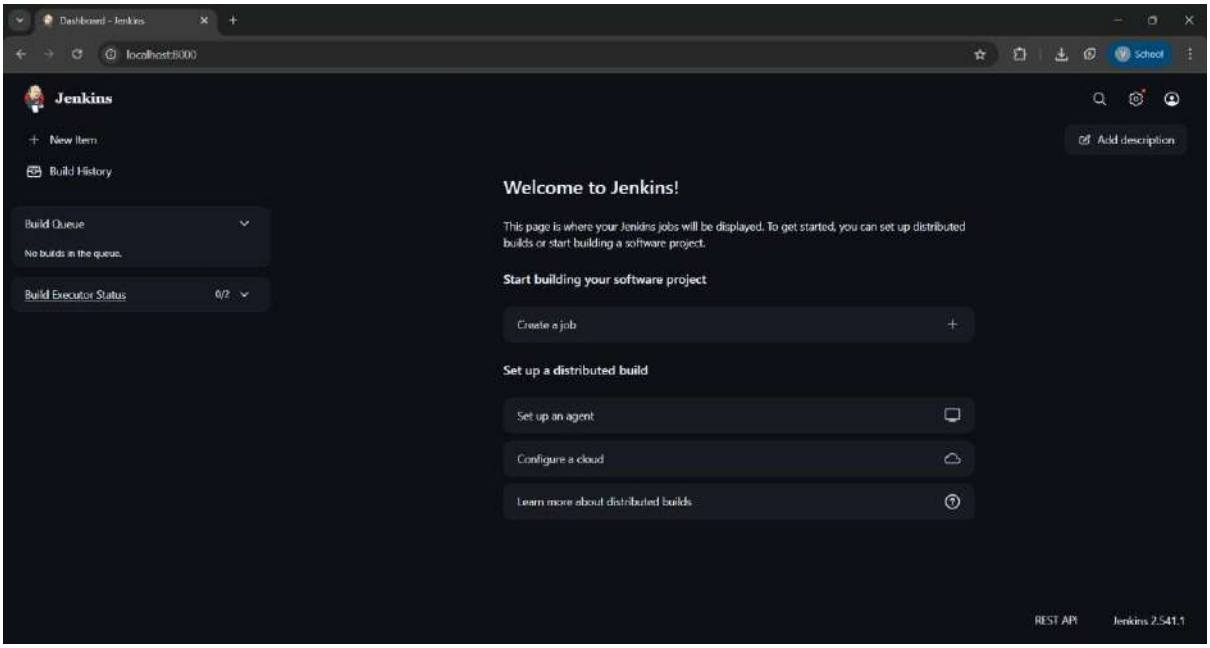
   With CI/CD:

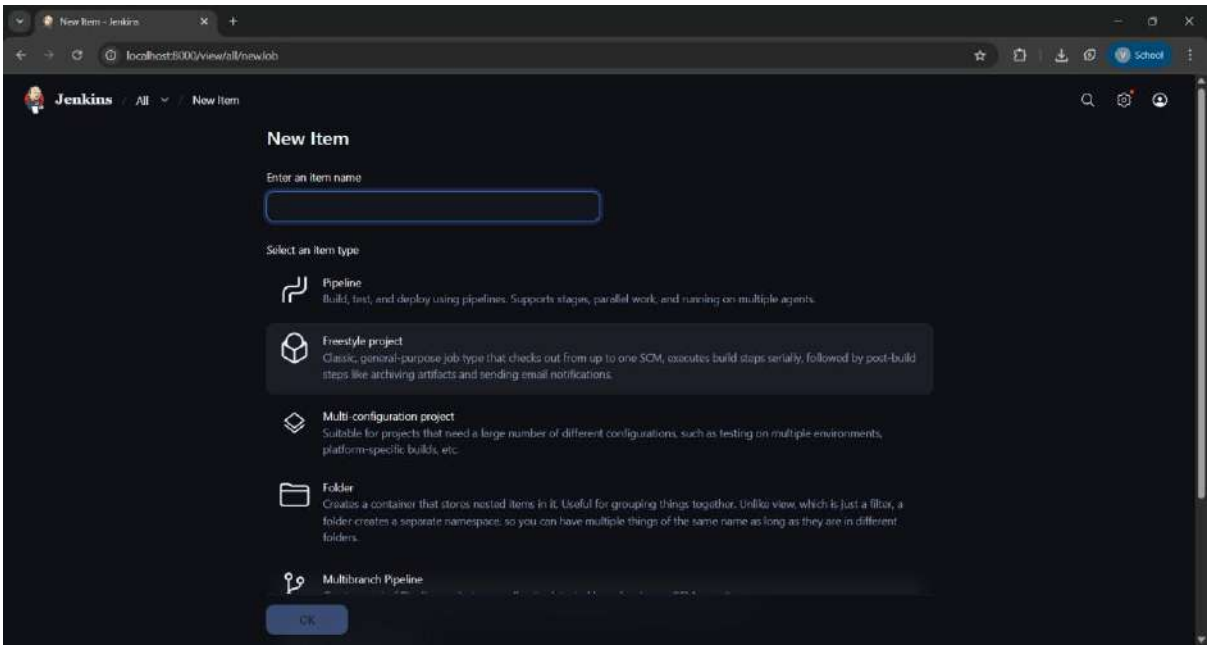
   - Code is pushed to GitHub
   - Jenkins **automatically builds** the project
   - Runs **tests**
   - Prepares it for **deployment**

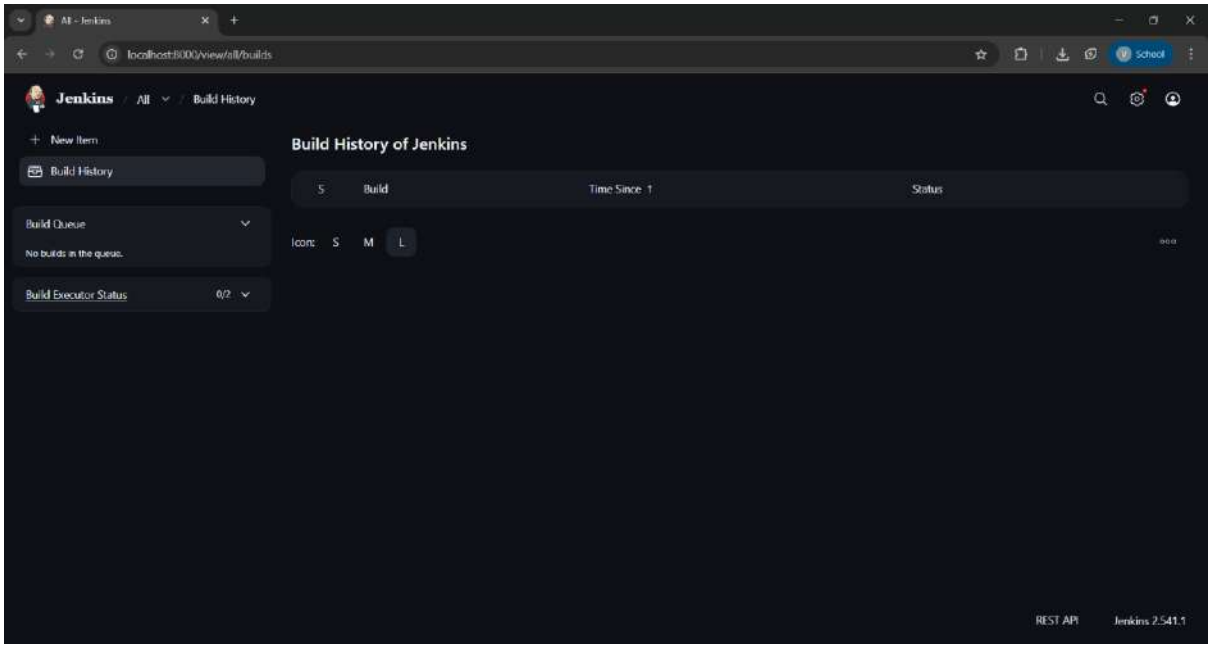
2. Install Jenkins

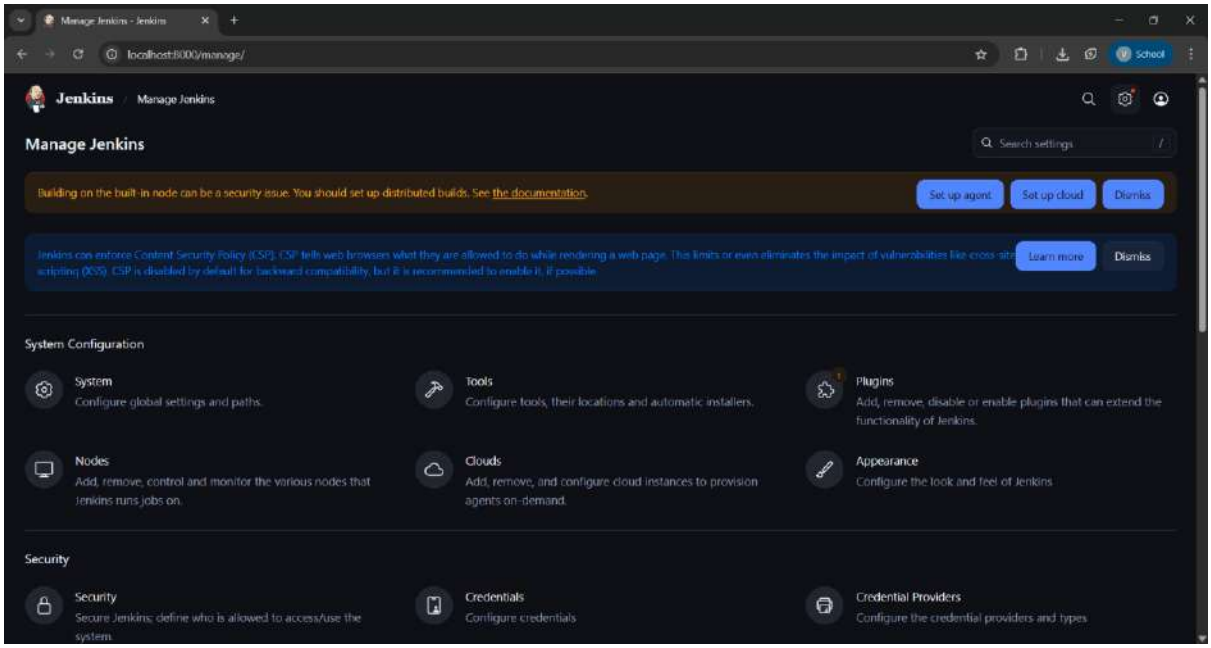
   Already installed in windows

3. Jenkins UI overview

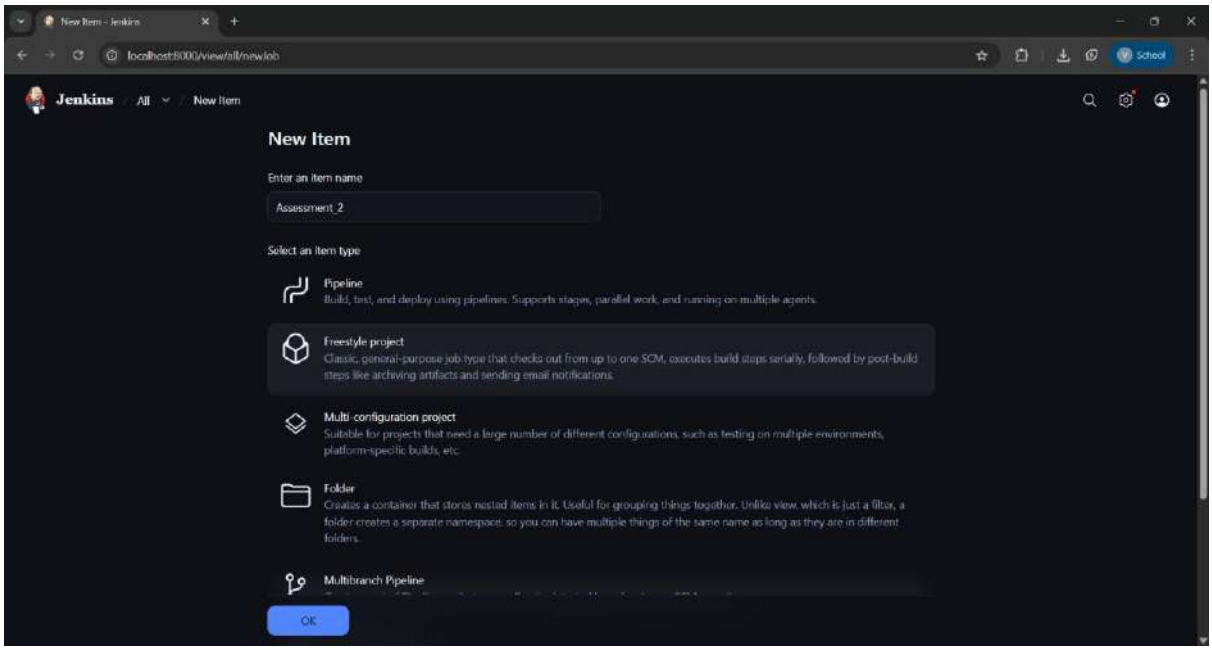




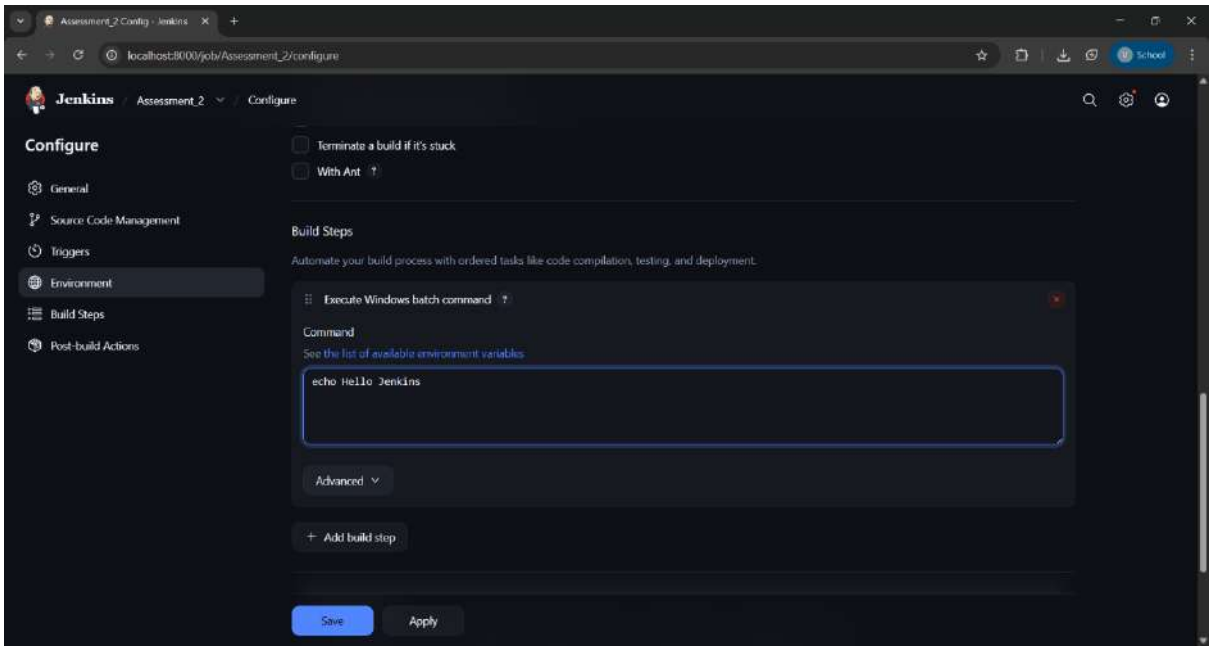

All - Jenkins
localhost:8000/view/all/builds
Jenkins / All / Build History
New Item
Build History
Build Queue
No builds in the queue.
Build Executor Status 0/2
Build History of Jenkins
S
Build
Time Since ↑
Status
Icon: S M L
REST API
Jenkins 2.541.1


Manage Jenkins - Jenkins
localhost:8000/manage/
Jenkins / Manage Jenkins
Manage Jenkins
Search settings
Building on the built-in node can be a security issue. You should set up distributed builds. See the documentation.
Set up agent
Set up cloud
Dismiss
Learn more
Dismiss
System Configuration
System
Configure global settings and paths.
Tools
Configure tools, their locations and automatic installers.
Plugins
Add, remove, disable or enable plugins that can extend the functionality of Jenkins.
Nodes
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.
Clouds
Add, remove, and configure cloud instances to provision agents on-demand.
Appearance
Configure the look and feel of Jenkins
Security
Security
Secure Jenkins; define who is allowed to access/use the system.
Credentials
Configure credentials
Credential Providers
Configure the credential providers and types

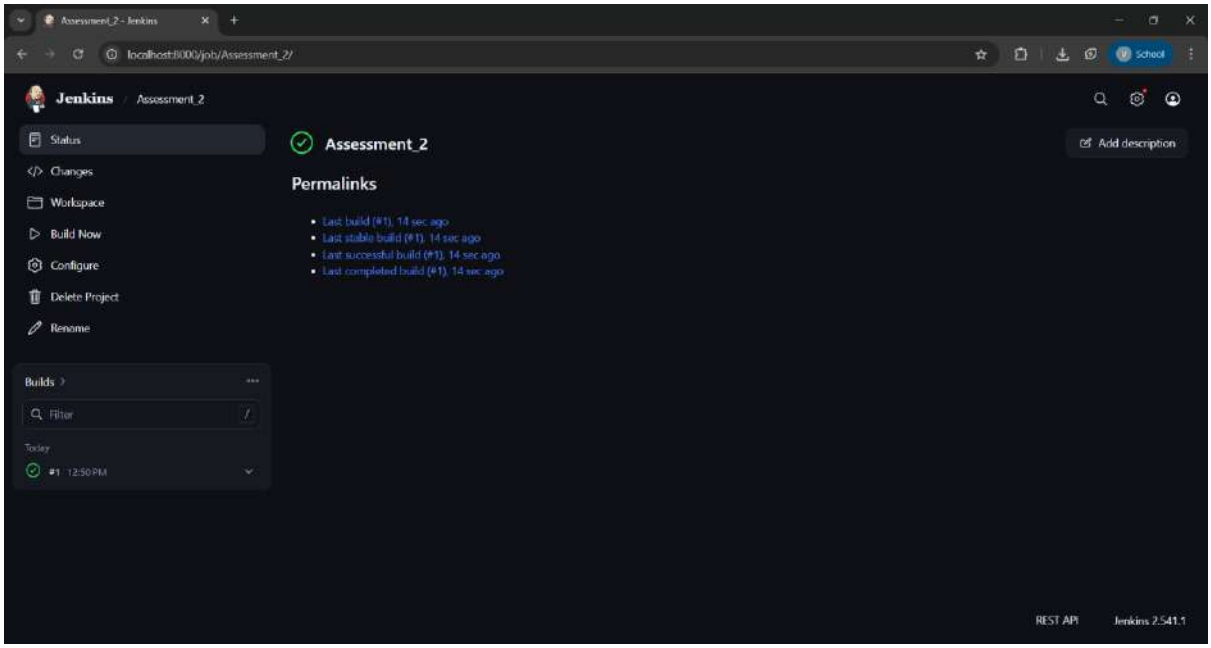
4. reate **Freestyle Job**



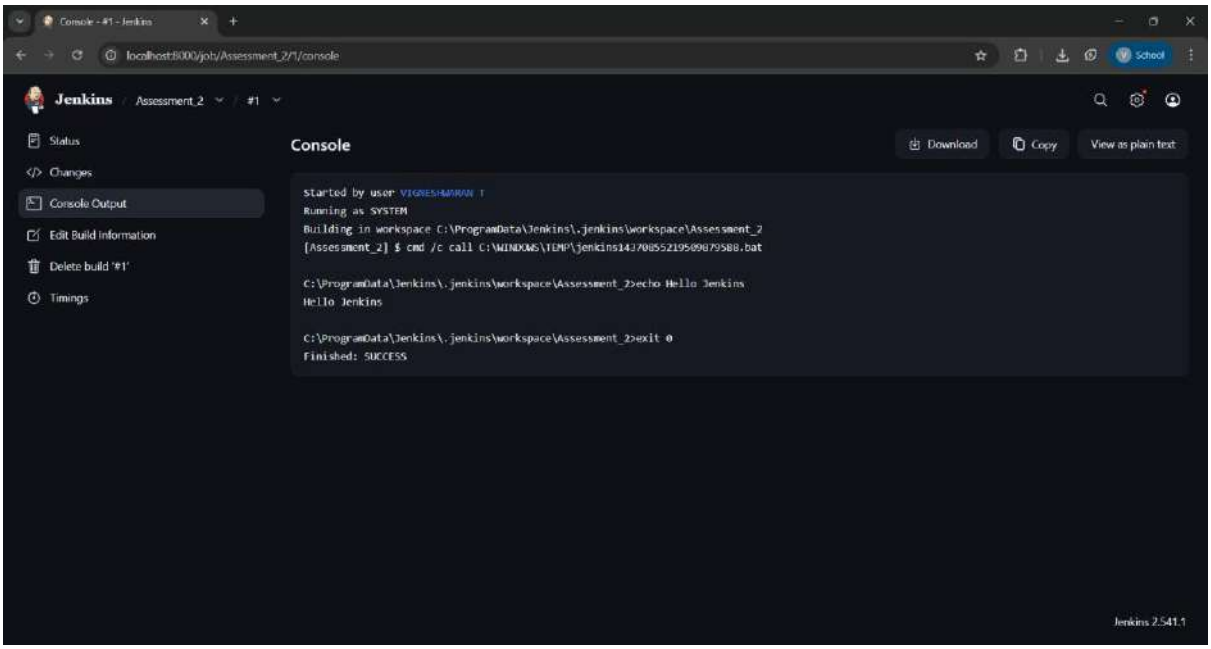
5. Add simple build step:
   - Windows: `echo Hello Jenkins`



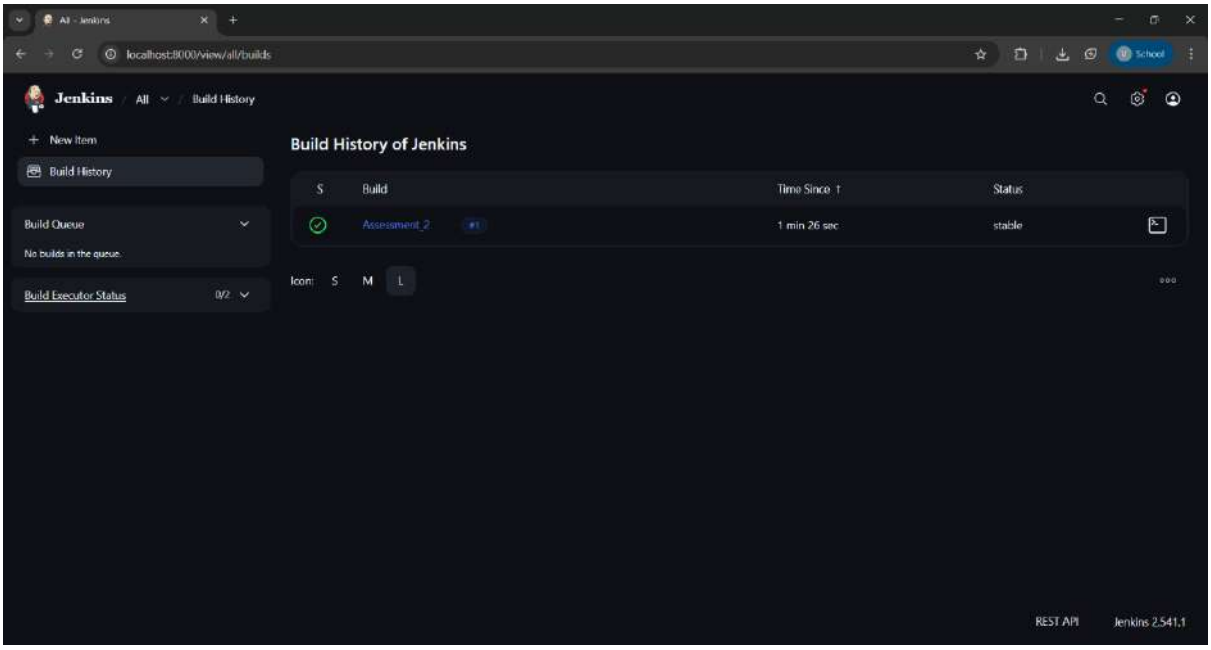
6. Run job manually



7. View:
   - Console Output



- Build History



Outcome

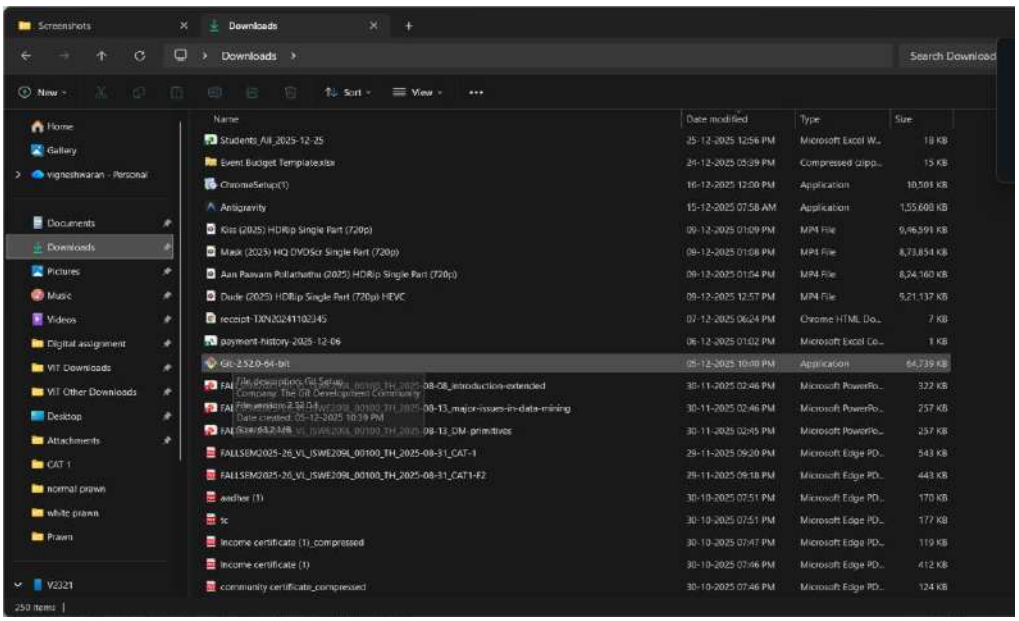
A learner can install Jenkins and run a basic job.

2. Learning Goal

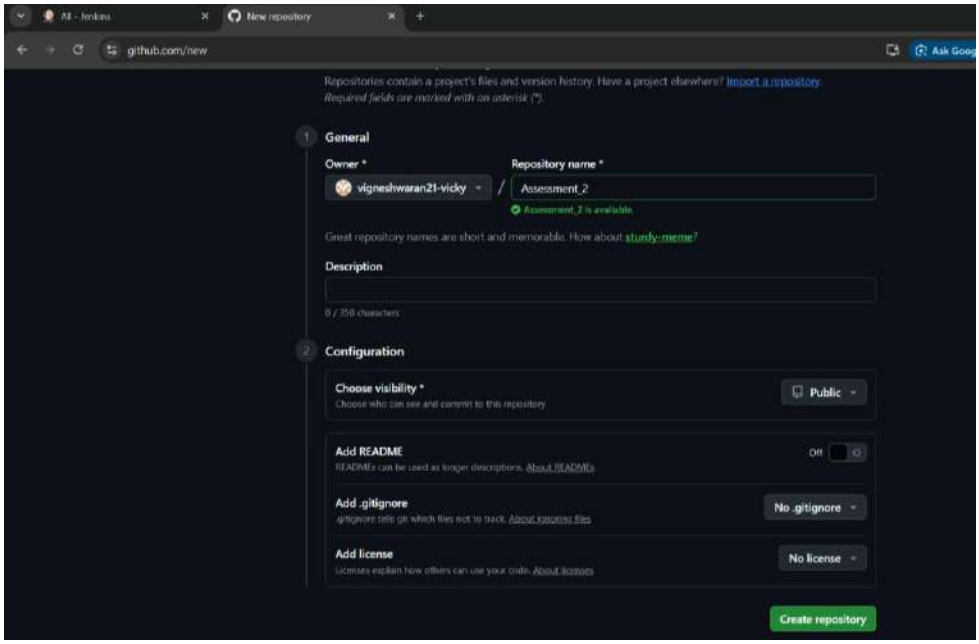
Work with **source code and build automation**

Stepwise Tasks

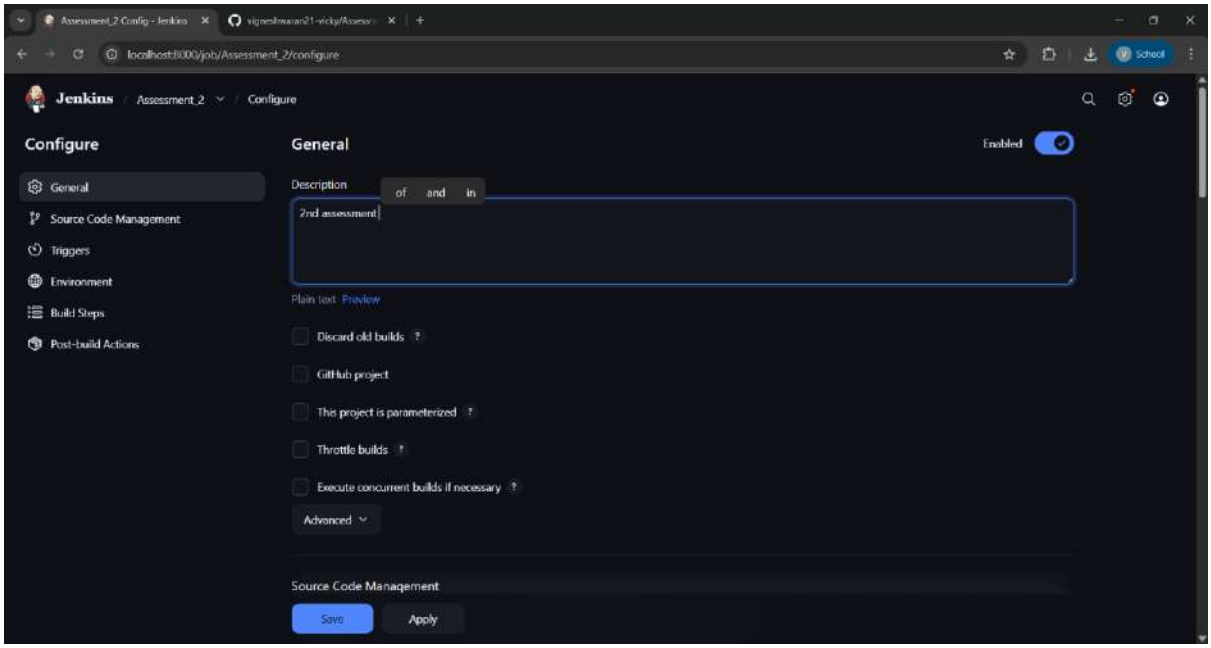
1. Install Git

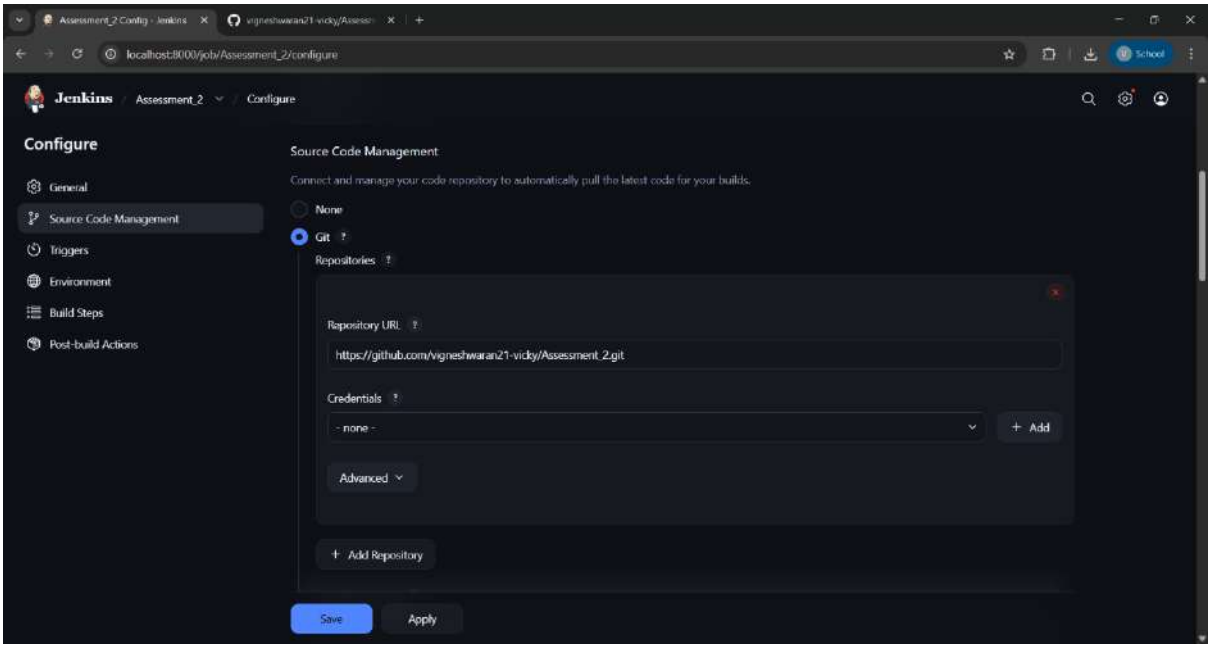


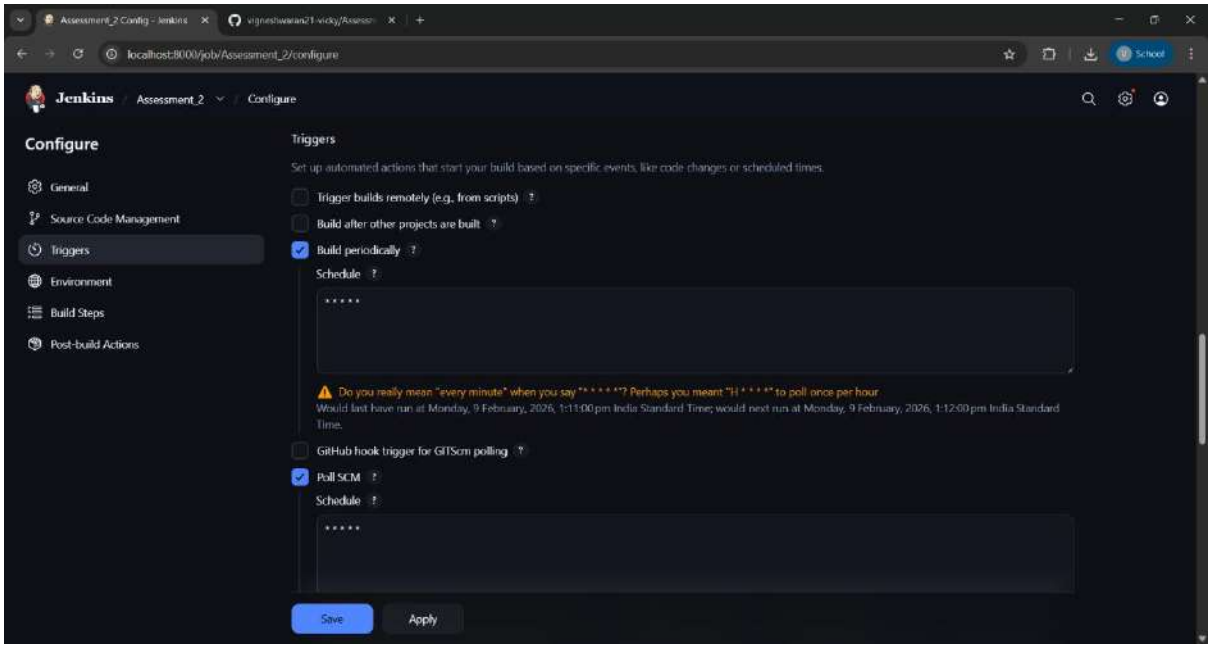
2. Create GitHub repository

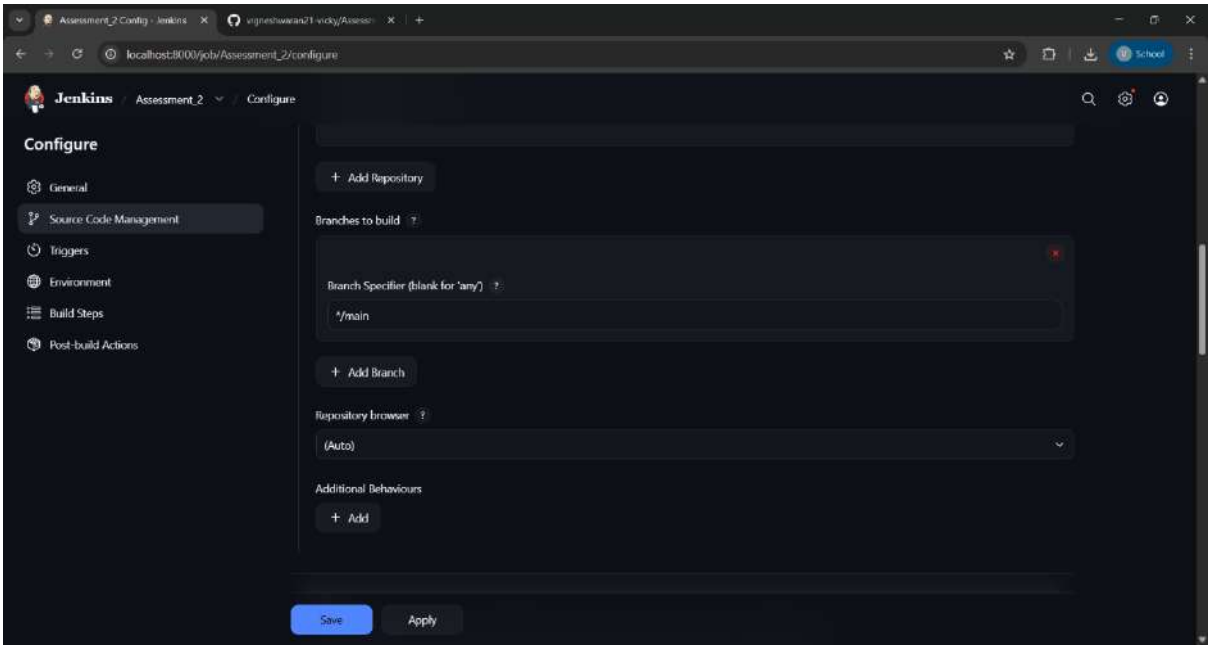


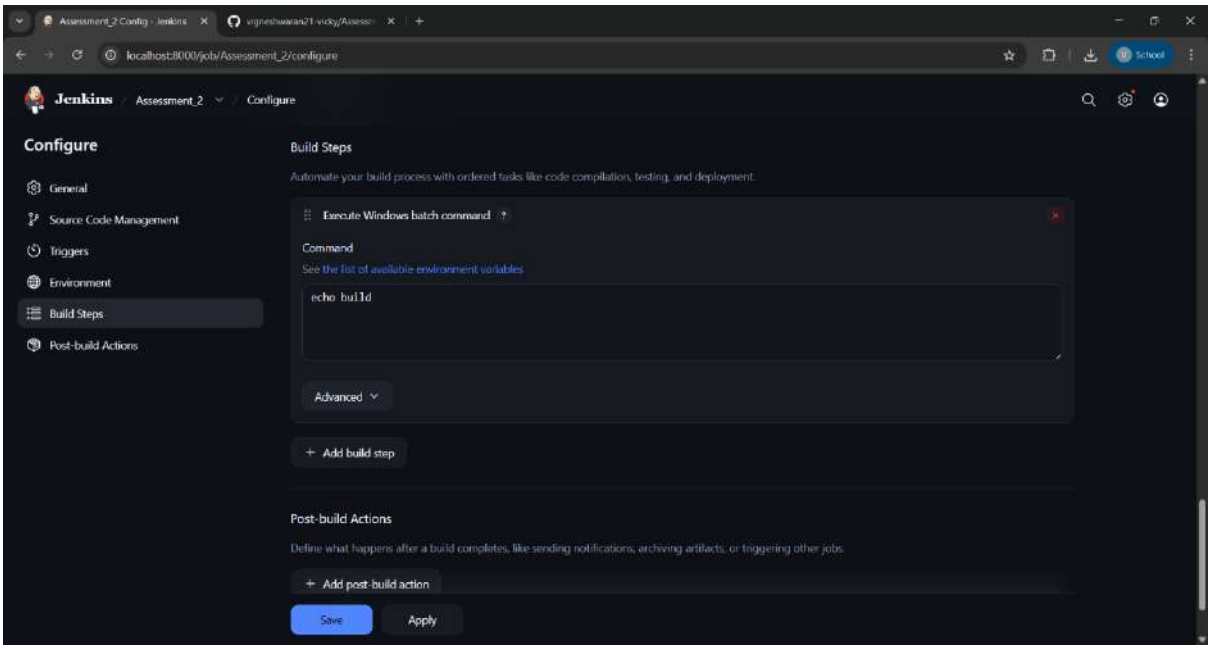
3. Configure **Git in Jenkins**

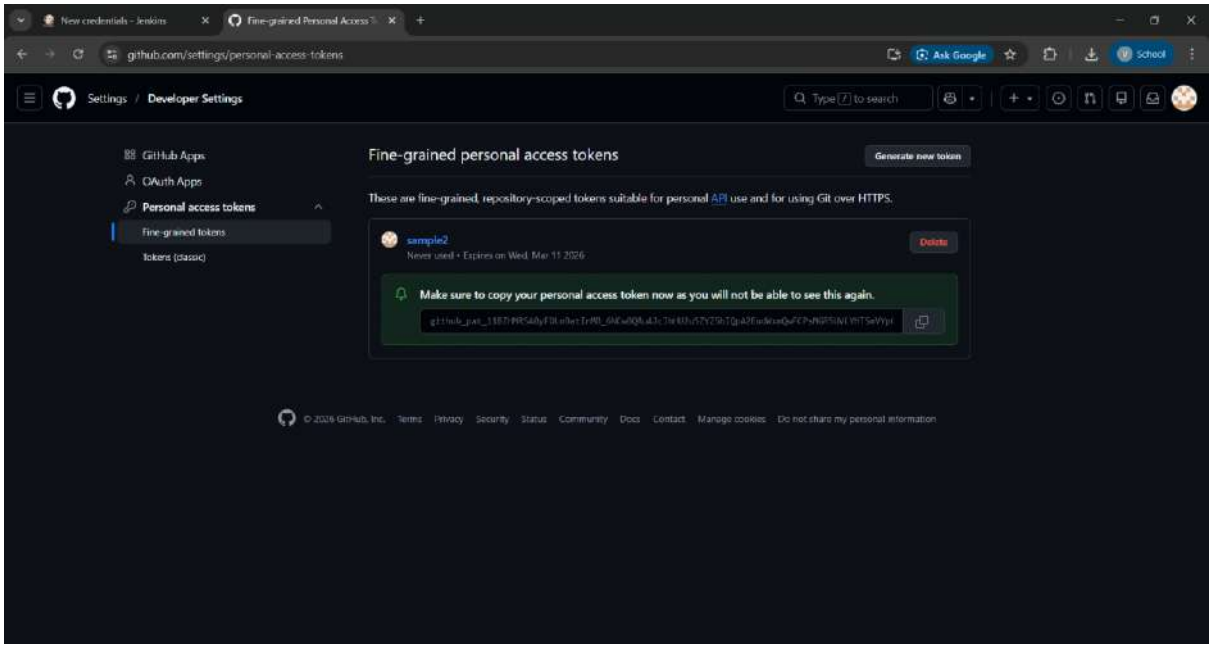


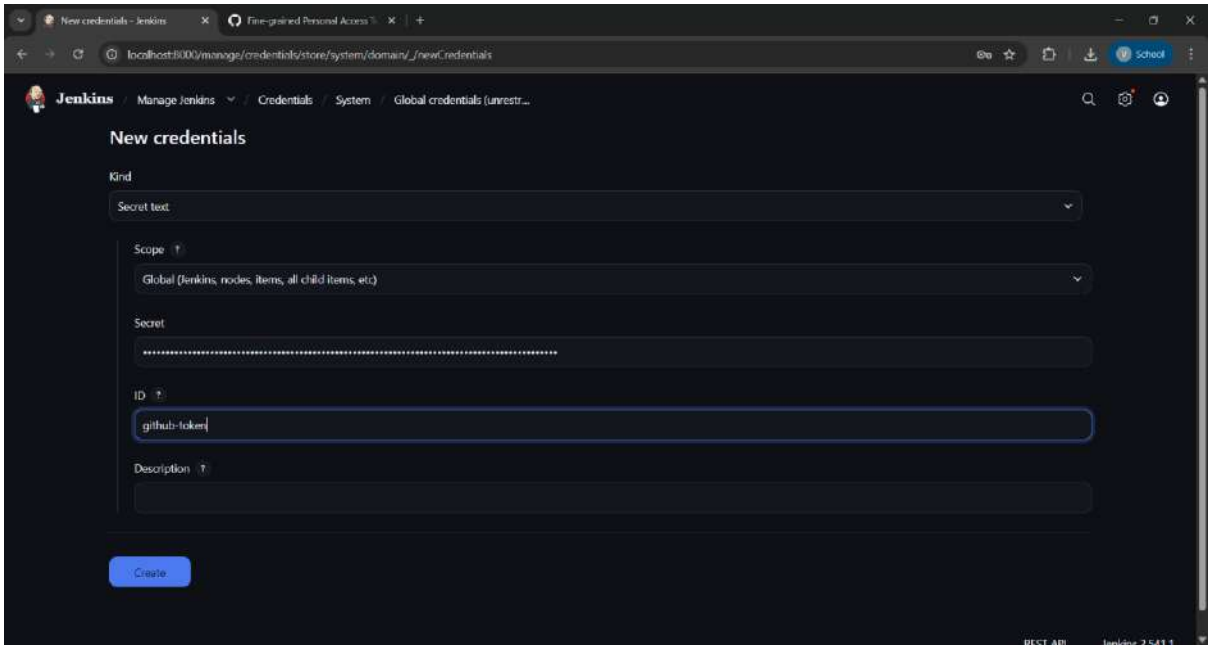

Assessment_2 Config - Jenkins
localhost:8000/job/Assessment_2/configure
Jenkins
Assessment_2
Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Source Code Management
Connect and manage your code repository to automatically pull the latest code for your builds.
None
Git
Repositories
Repository URL
https://github.com/vigneshwaran21-vicky/Assessment_2.git
Credentials
- none -
+ Add
Advanced
+ Add Repository
Save
Apply


Assessment_2 Config - Jenkins
localhost:8000/job/Assessment_2/configure
Jenkins
Assessment_2
Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Triggers
Set up automated actions that start your build based on specific events, like code changes or scheduled times.
Trigger builds remotely (e.g., from scripts)
Build after other projects are built
Build periodically
Schedule
* * * * *
Do you really mean "every minute" when you say "* * * * *"? Perhaps you meant "H * * * *" to poll once per hour
Would last have run at Monday, 9 February, 2026, 1:11:00 pm India Standard Time; would next run at Monday, 9 February, 2026, 1:12:00 pm India Standard Time.
GitHub hook trigger for GITScm polling
Poll SCM
Schedule
* * * * *
Save
Apply

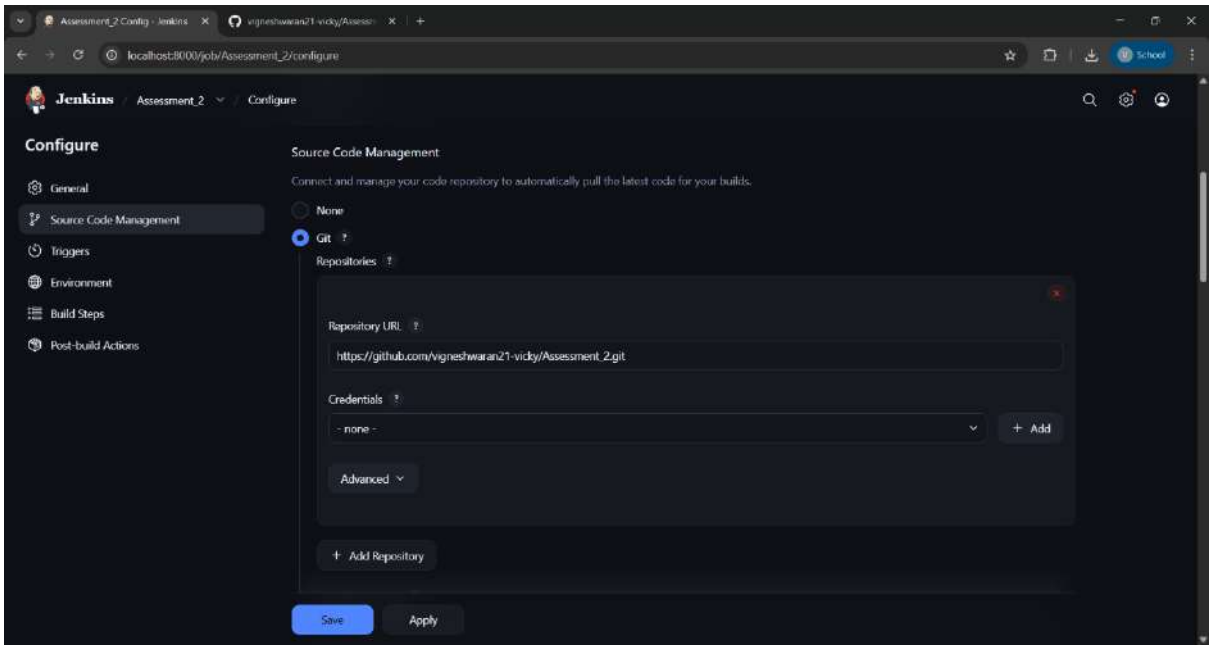

Assessment_2 Config - Jenkins
localhost:8000/job/Assessment_2/configure
Jenkins
Assessment_2
Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
+ Add Repository
Branches to build
Branch Specifier (blank for 'any')
*/main
+ Add Branch
Repository browser
(Auto)
Additional Behaviours
+ Add
Save
Apply

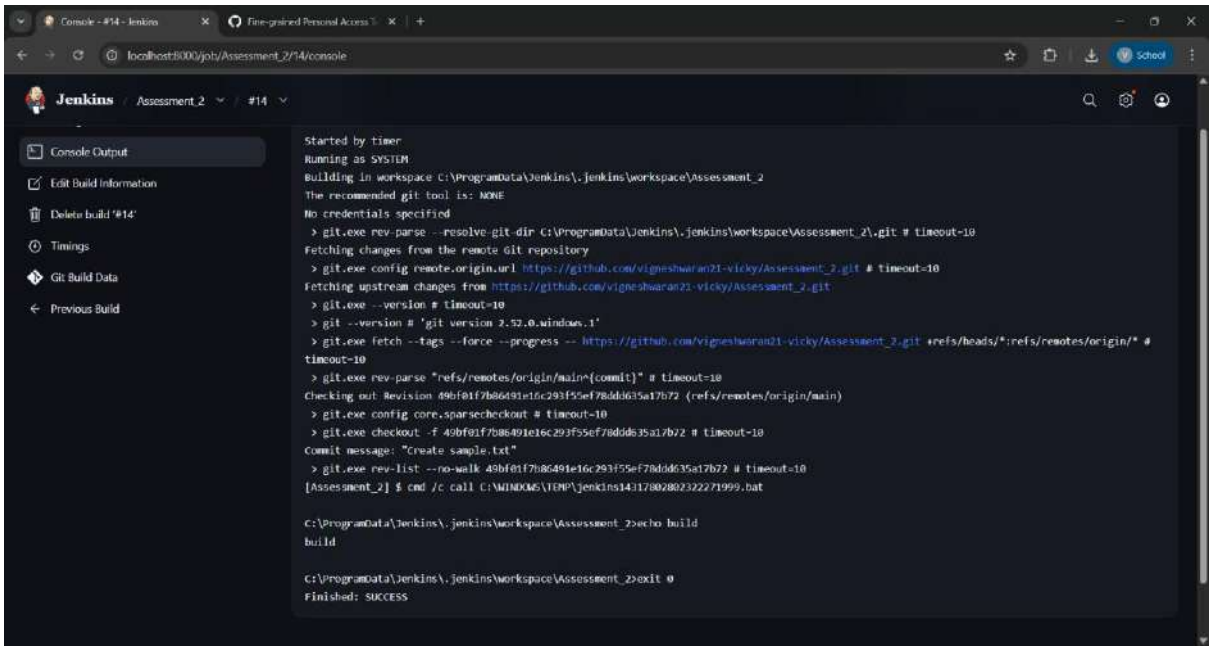

Assessment_2 Config - Jenkins
localhost:8000/job/Assessment_2/configure
Jenkins
Assessment_2
Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Build Steps
Automate your build process with ordered tasks like code compilation, testing, and deployment.
Execute Windows batch command
Command
See the list of available environment variables
echo build
Advanced
+ Add build step
Post-build Actions
Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.
+ Add post-build action
Save
Apply

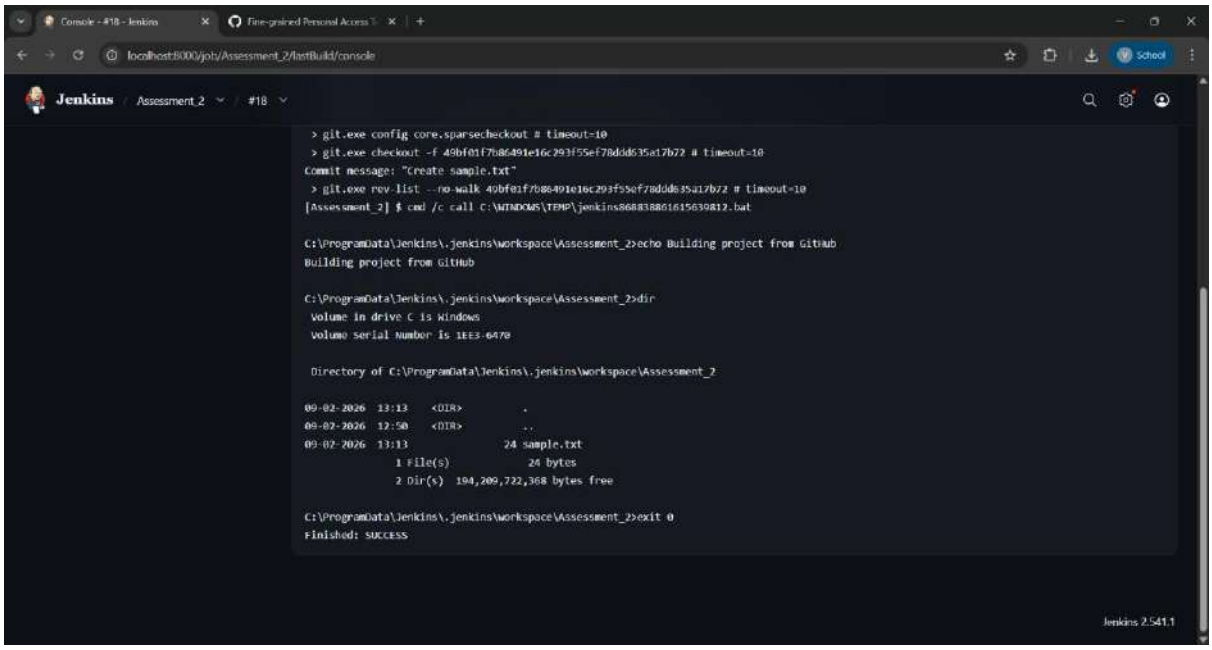

New credentials - Jenkins
Fine-grained Personal Access T
github.com/settings/personal-access-tokens
Ask Google
School
Settings / Developer Settings
Type / to search
GitHub Apps
OAuth Apps
Personal access tokens
Fine-grained tokens
Tokens (classic)
Fine-grained personal access tokens
Generate new token
These are fine-grained, repository-scoped tokens suitable for personal API use and for using Git over HTTPS.
sample2
Never used • Expires on Wed, Mar 11 2026
Delete
Make sure to copy your personal access token now as you will not be able to see this again.
© 2026 GitHub, Inc. Terms Privacy Security Status Community Docs Contact Manage cookies Do not share my personal information


New credentials - Jenkins
Fine-grained Personal Access T
localhost:8000/manage/credentials/store/system/domain/_/newCredentials
School
Jenkins
Manage Jenkins
Credentials
System
Global credentials (unrestr...
New credentials
Kind
Secret text
Scope
Global (Jenkins, nodes, items, all child items, etc)
Secret
ID
github-token
Description
Create

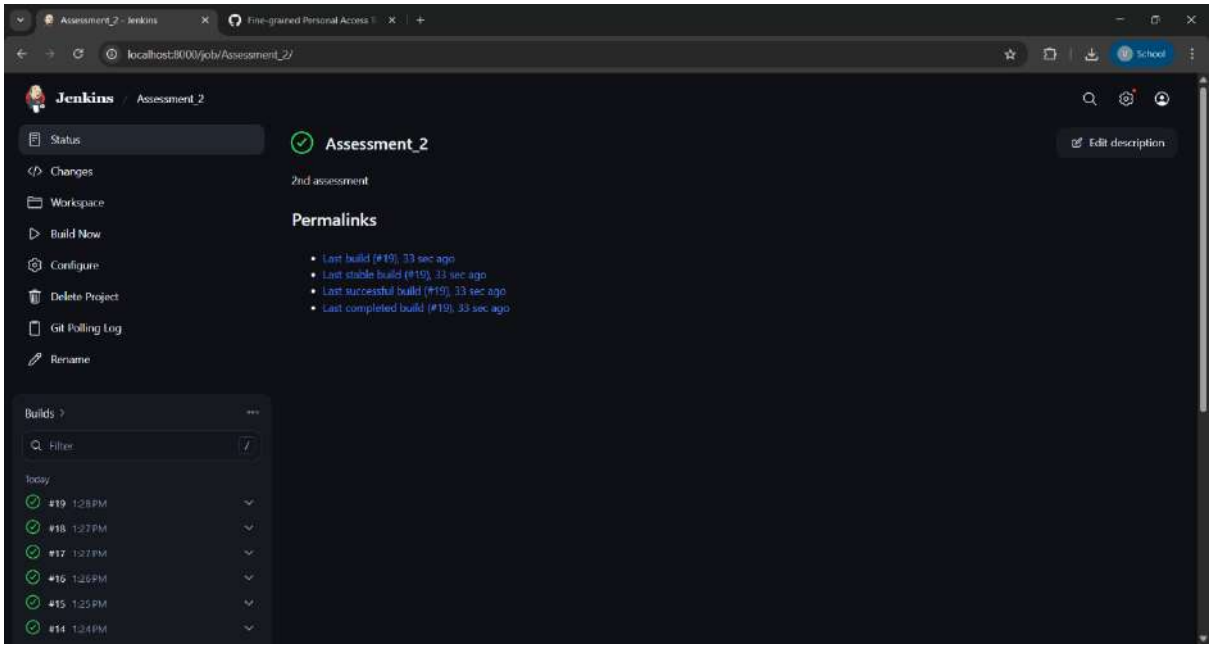
4. Create Freestyle Job:
    - Source Code Management → Git



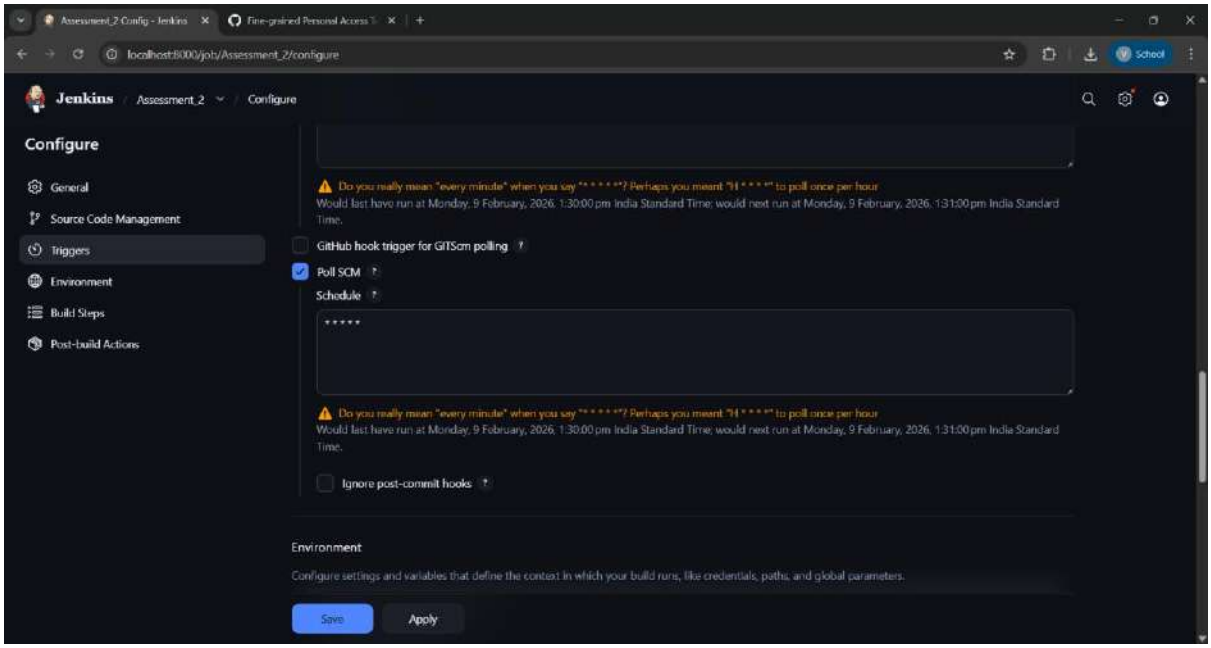




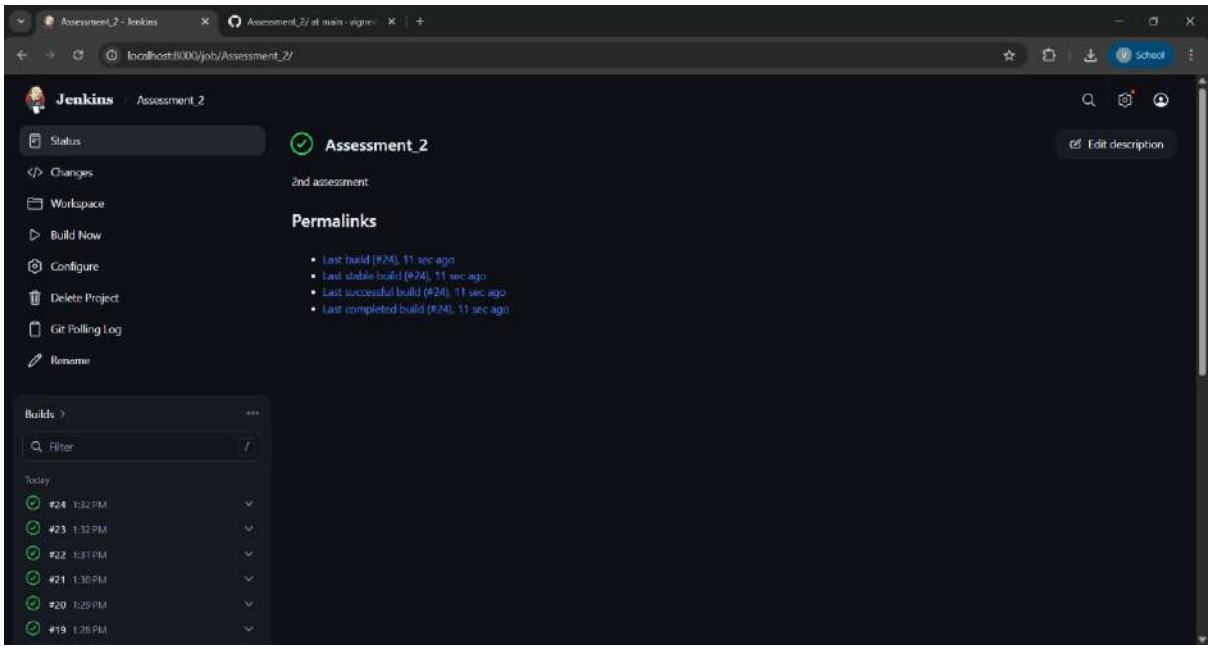
5. Add build step:
   - Windows batch / Shell

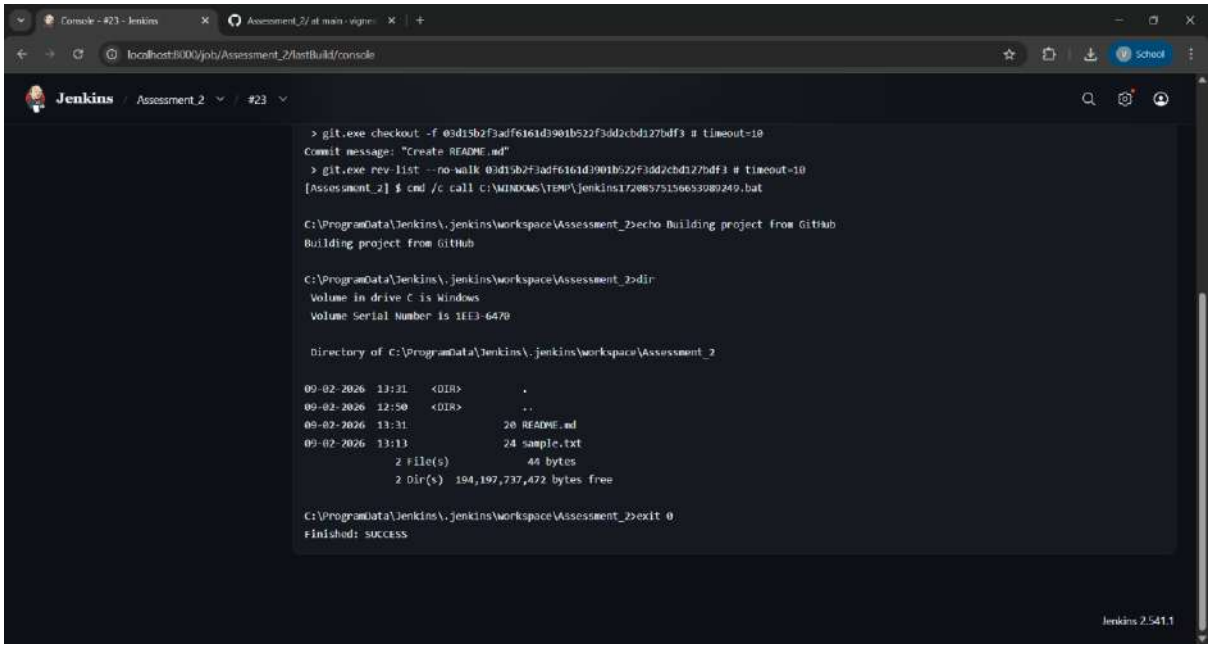


6. Poll SCM / Webhook trigger



7. Build automatically on code change





Outcome

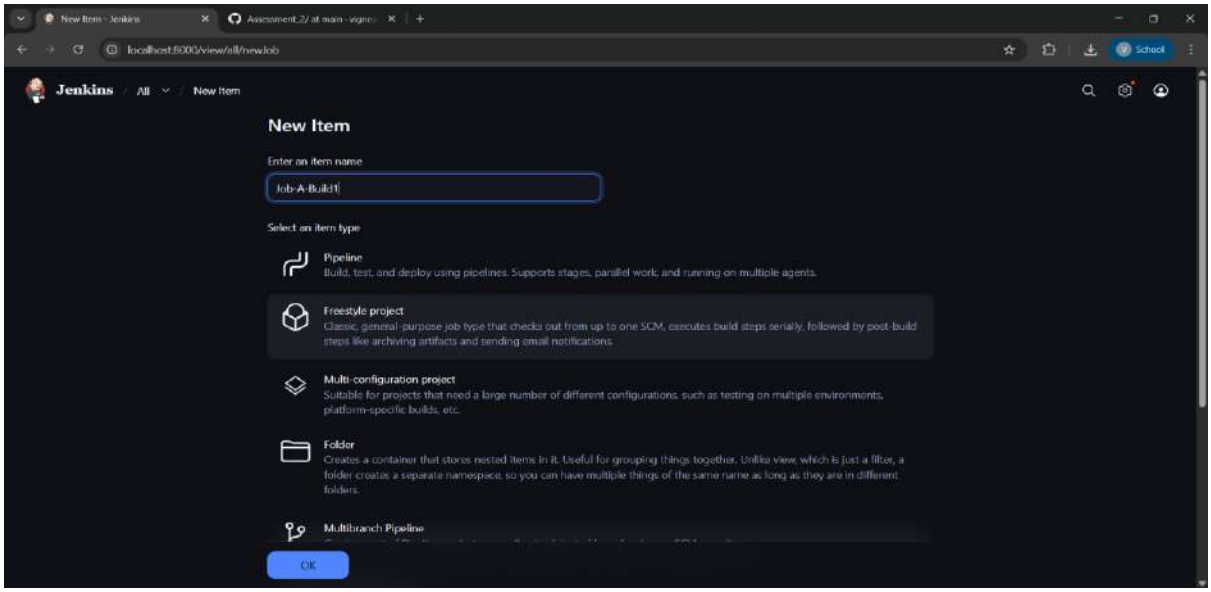
Learner connects Jenkins with GitHub.

3. Learning Goal

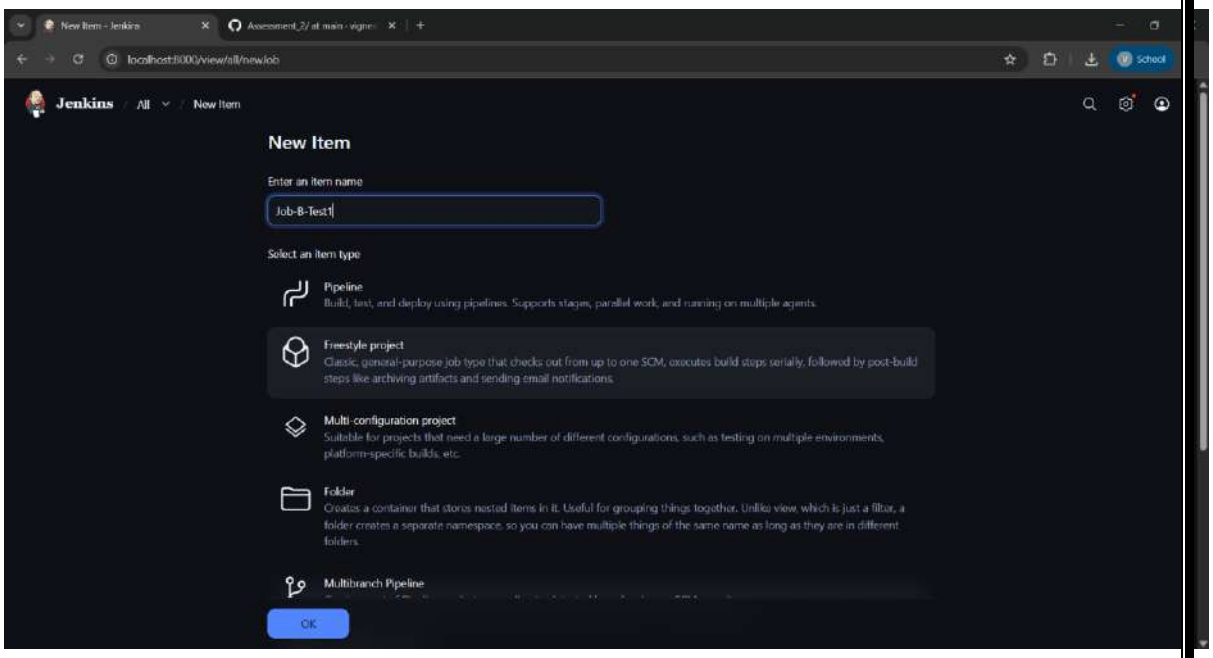
Understand **job flow visually**

Stepwise Tasks

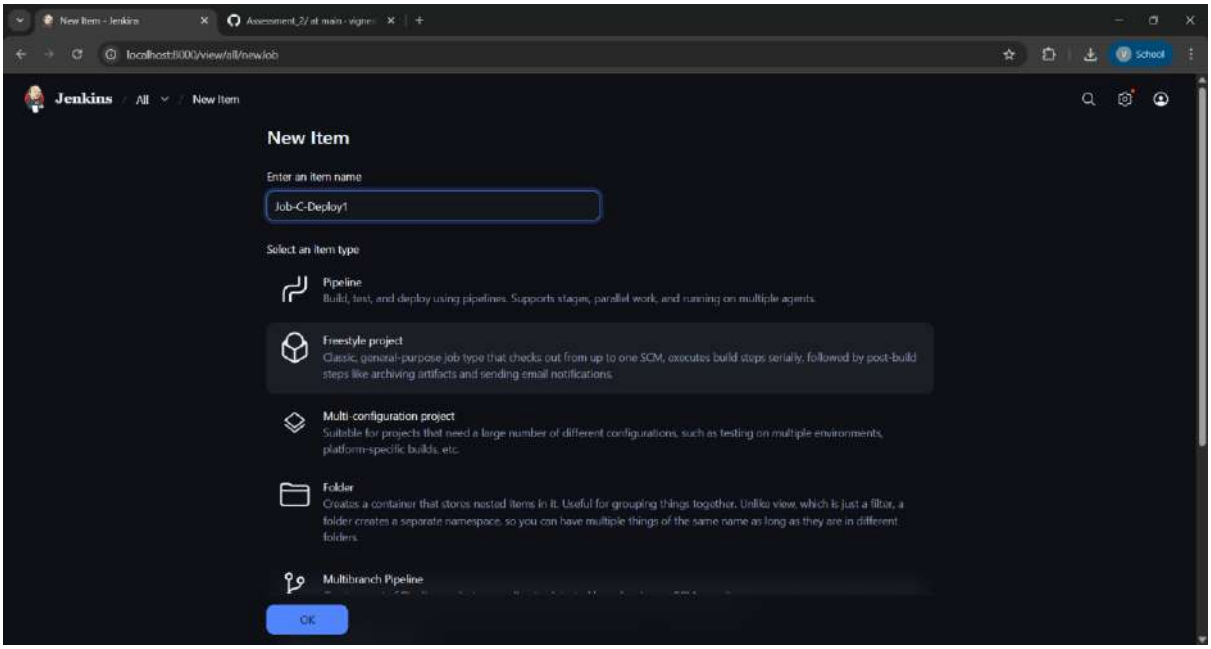
1. Create multiple Freestyle jobs:
    - Job-A (Build)

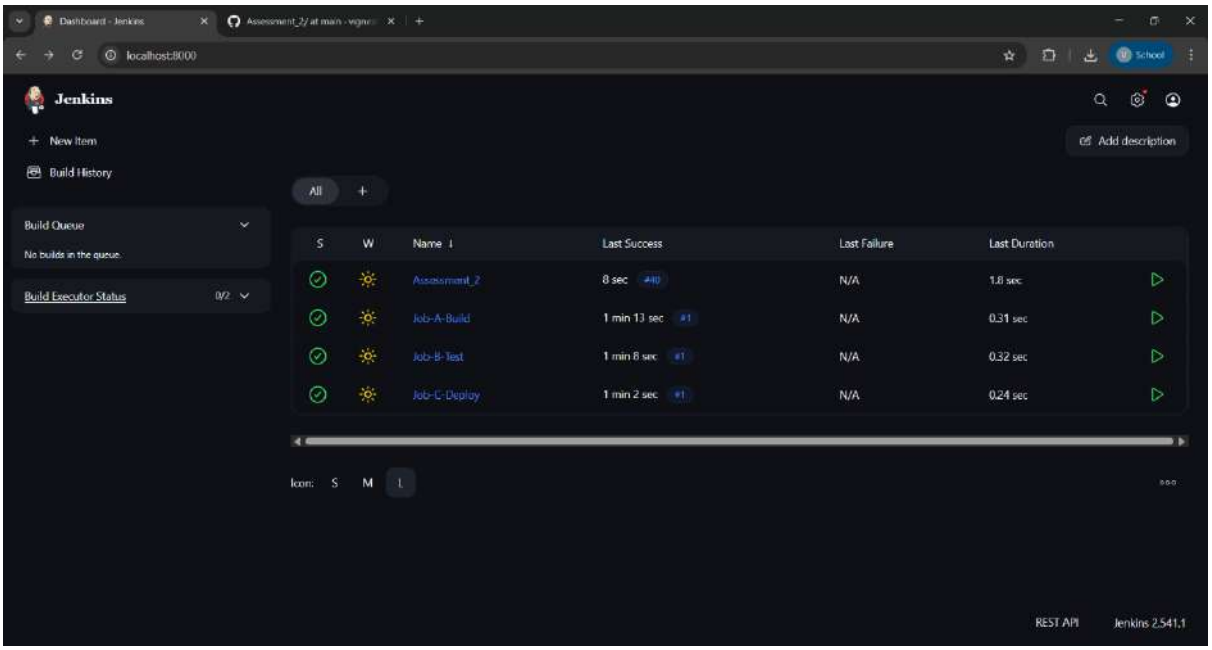


- Job-B (Test)

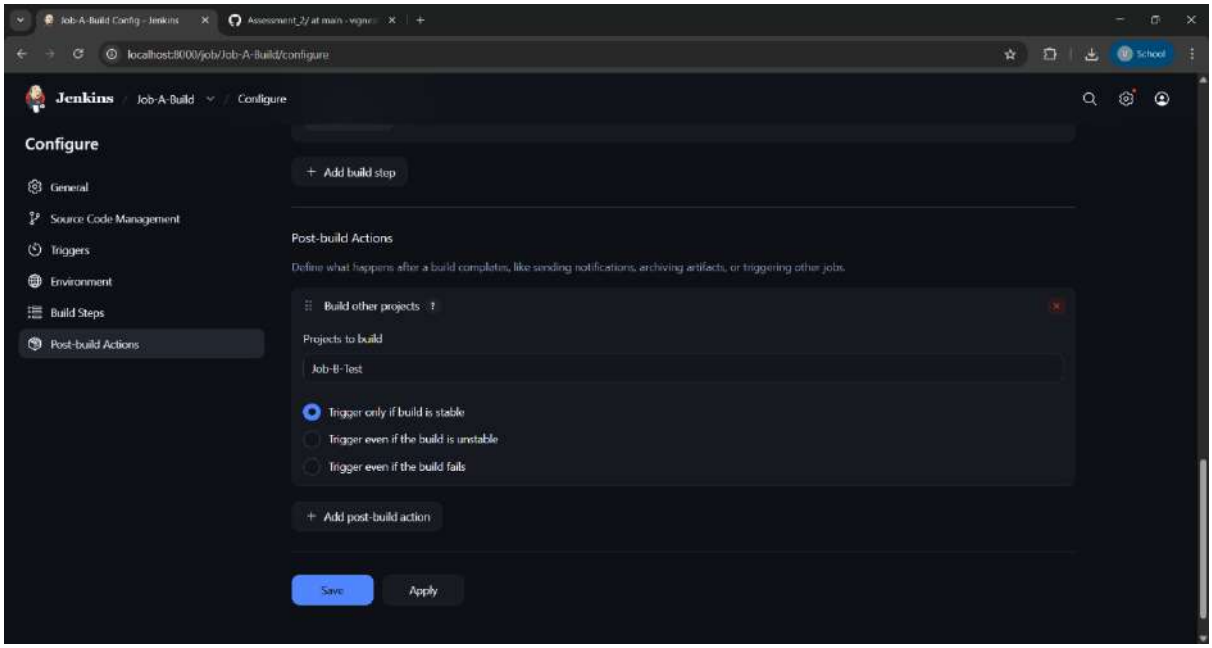


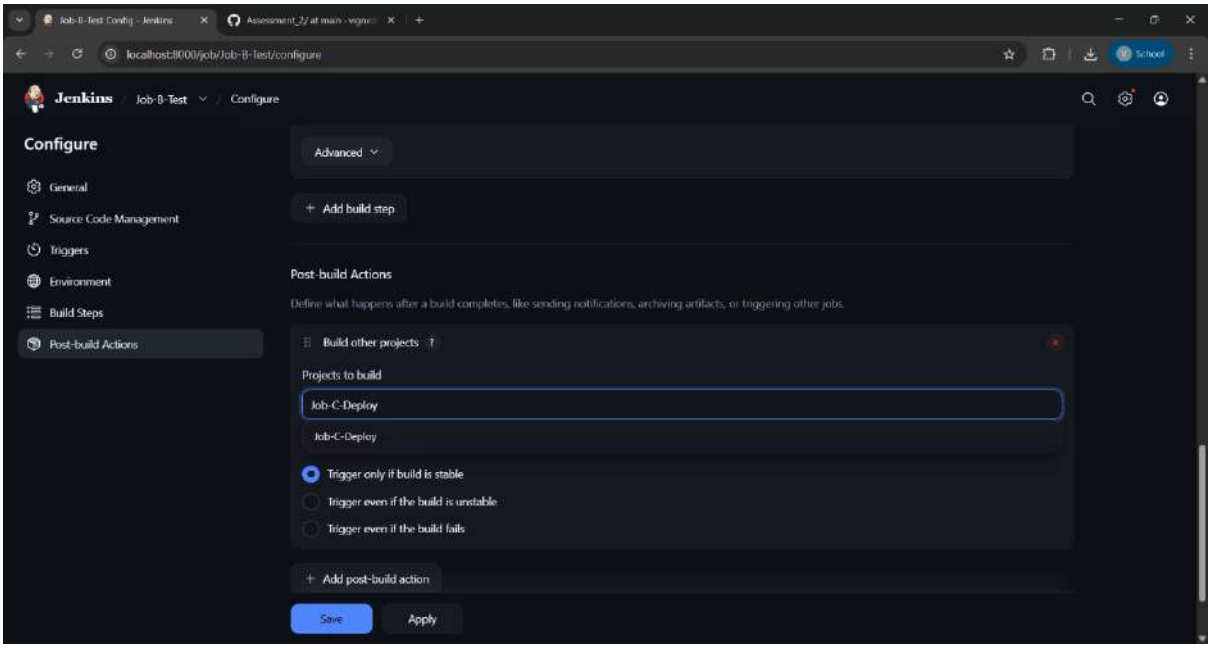
- Job-C (Deploy)



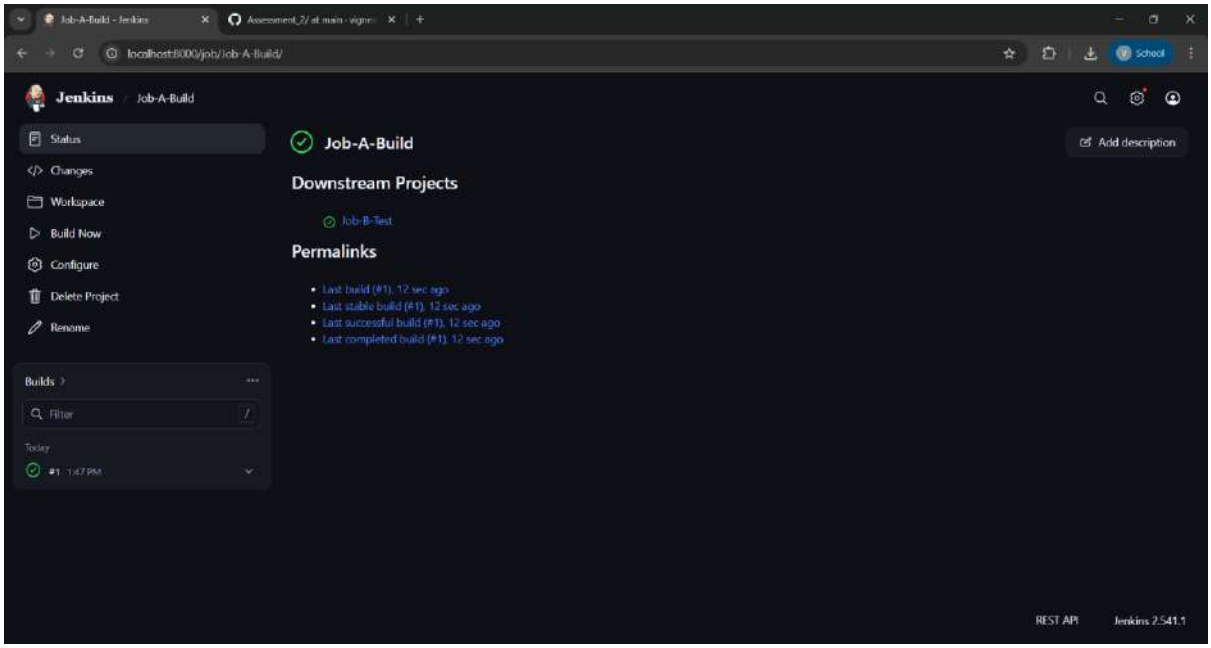


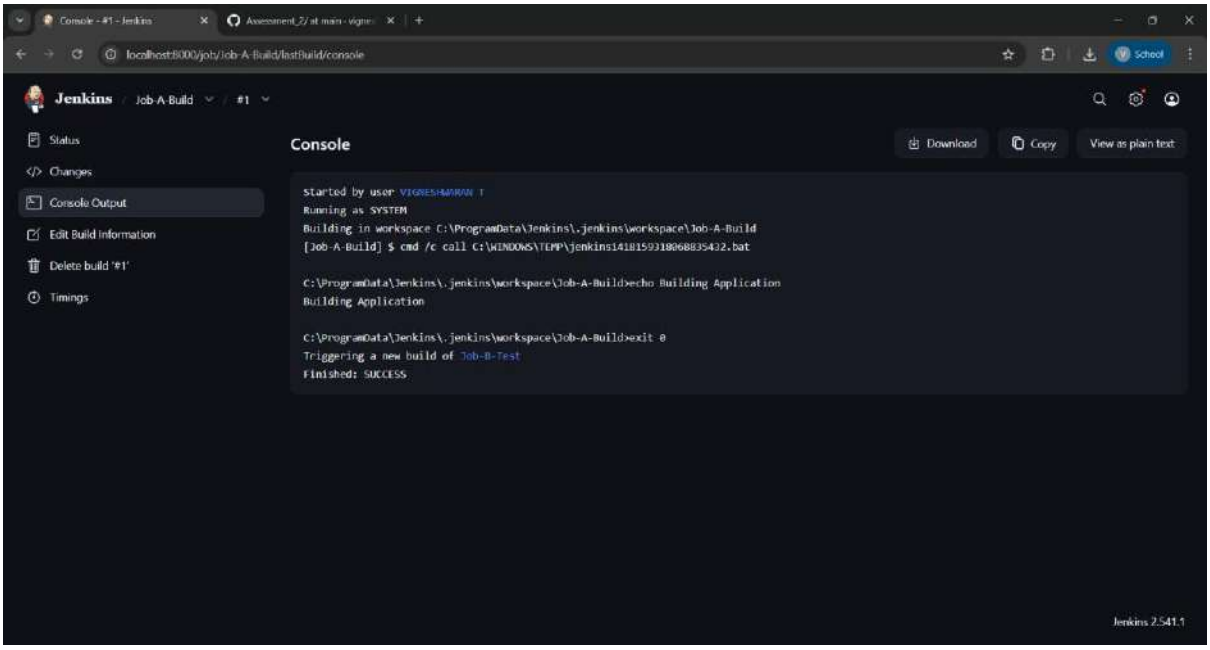
2. Use:
   - Post-build action → Trigger other projects

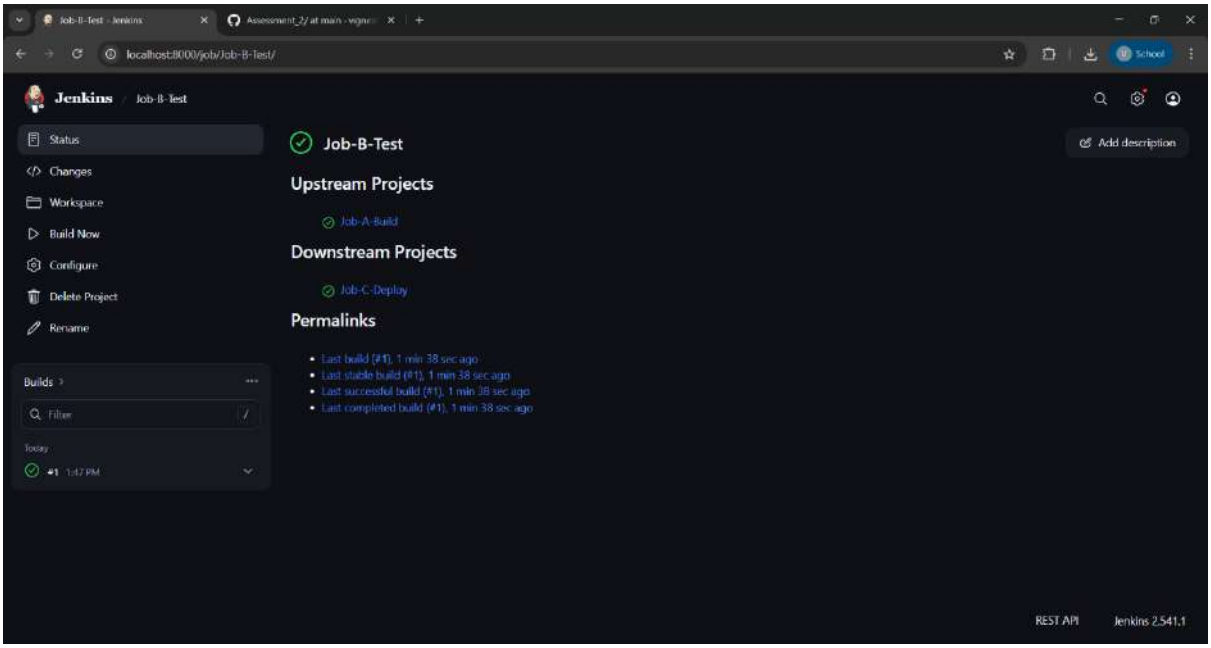


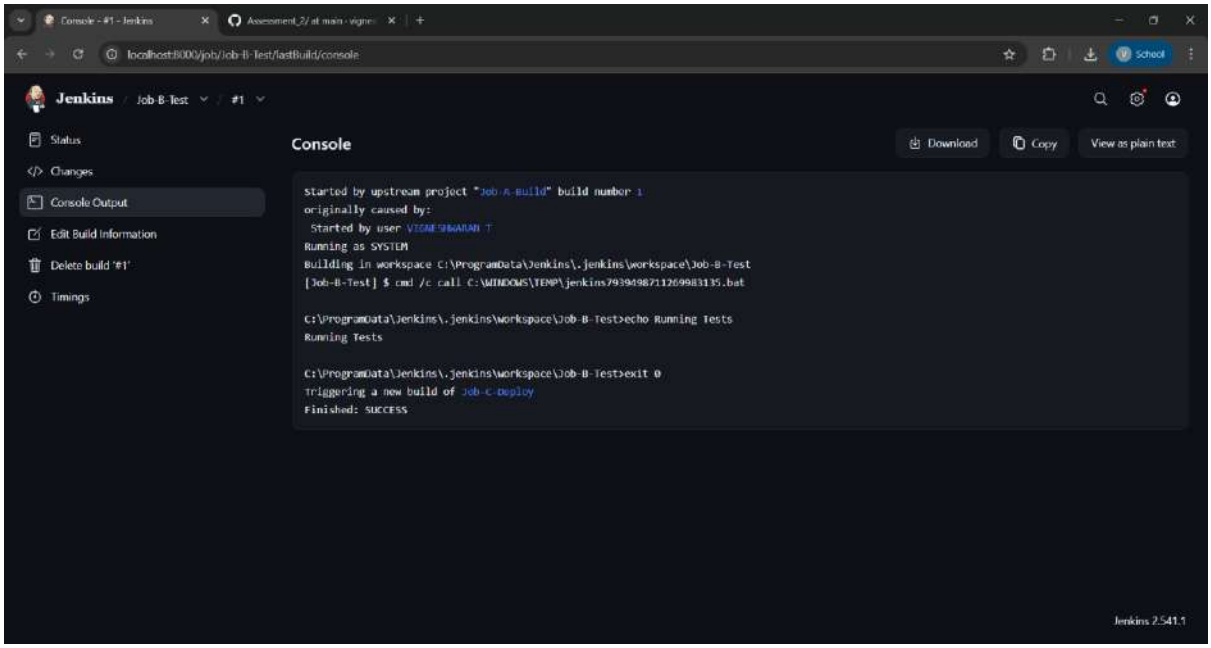


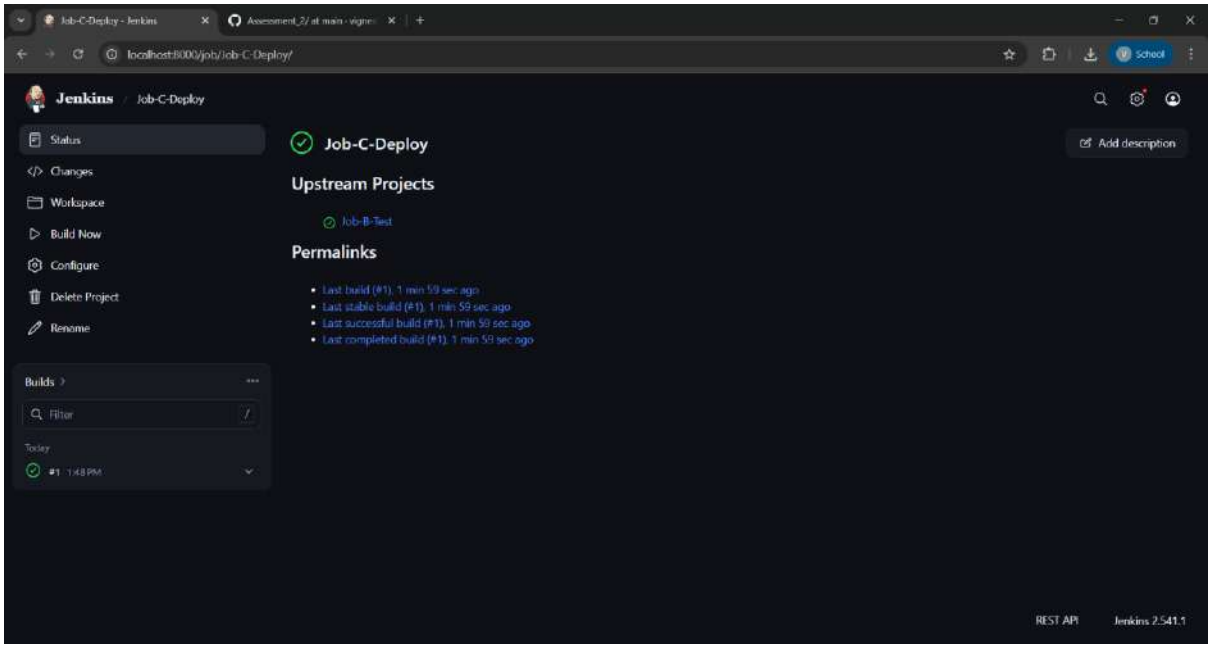
3. View job chaining visually

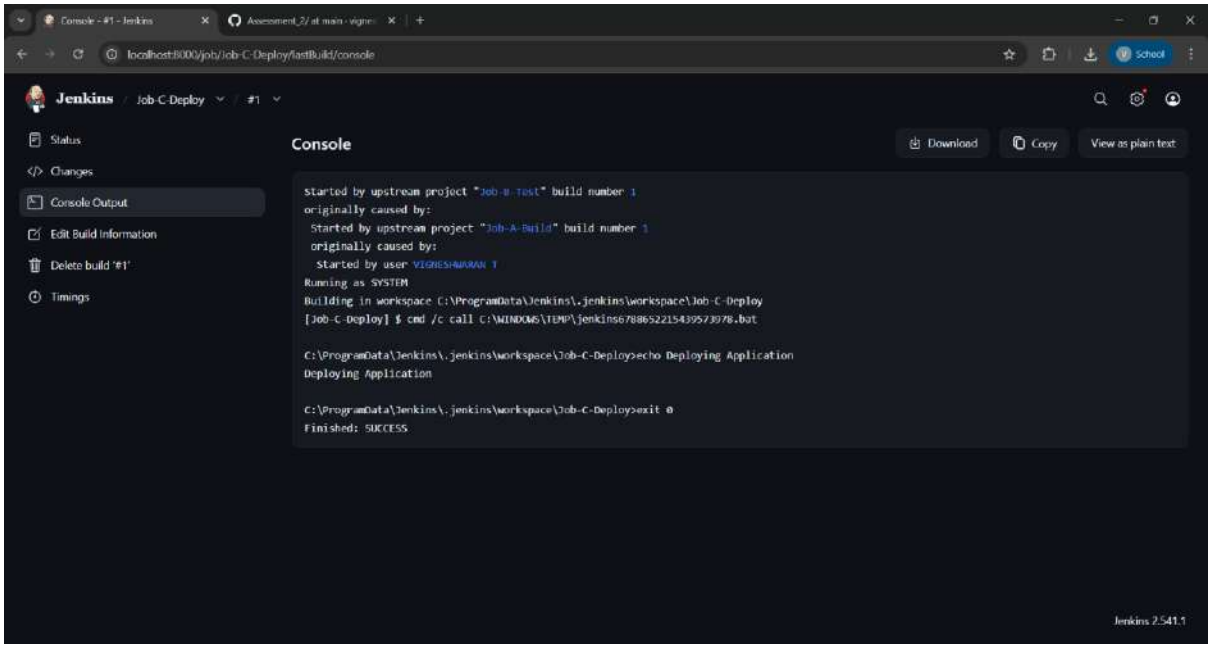



Console - #1 - Jenkins
Assessment_2/ at main - vigne
localhost:8000/job/Job-A-Build/lastBuild/console
Jenkins
Job-A-Build
#1
Status
Changes
Console Output
Edit Build Information
Delete build '#1'
Timings
Console
Download
Copy
View as plain text
Started by user VIGNESHWARAN T
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\.jenkins\workspace\Job-A-Build
[Job-A-Build] $ cmd /c call C:\WINDOWS\TEMP\jenkins14181593180688354 32.bat
C:\ProgramData\Jenkins\.jenkins\workspace\Job-A-Build>echo Building Application
Building Application
C:\ProgramData\Jenkins\.jenkins\workspace\Job-A-Build>exit 0
Triggering a new build of Job-B-Test
Finished: SUCCESS
Jenkins 2.541.1

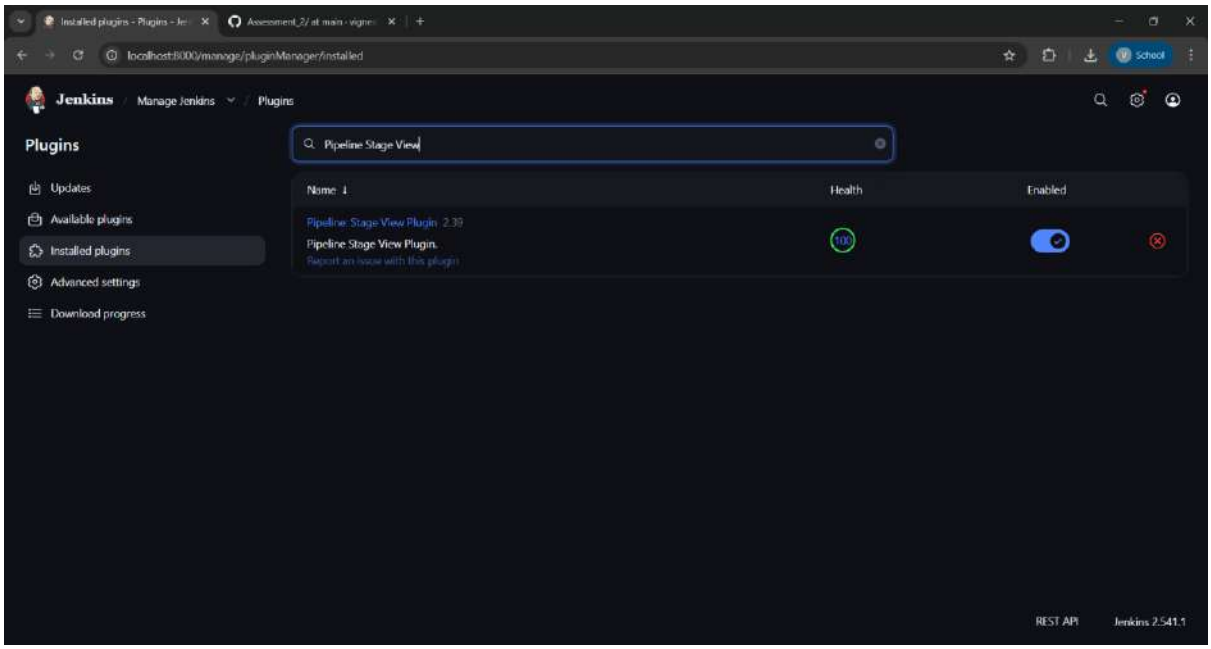

Job-B-Test - Jenkins
Assessment_2/ at main - vigne
localhost:8000/job/Job-B-Test/
Jenkins
Job-B-Test
Status
Changes
Workspace
Build Now
Configure
Delete Project
Rename
Builds
Filter
Today
#1 1:47 PM
Job-B-Test
Add description
Upstream Projects
Job-A-Build
Downstream Projects
Job-C-Deploy
Permalinks
Last build (#1), 1 min 38 sec ago
Last stable build (#1), 1 min 38 sec ago
Last successful build (#1), 1 min 38 sec ago
Last completed build (#1), 1 min 38 sec ago
REST API
Jenkins 2.541.1


Console - #1 - Jenkins
Assessment_2/ at main · vigne
localhost:8000/job/Job-B-Test/lastBuild/console
School
Jenkins / Job-B-Test / #1
Status
Changes
Console Output
Edit Build Information
Delete build '#1'
Timings
Console
Download
Copy
View as plain text
Started by upstream project "Job-A-Build" build number 1
originally caused by:
Started by user VIGNESHWARAN T
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\.jenkins\workspace\Job-B-Test
[Job-B-Test] $ cmd /c call C:\WINDOWS\TEMP\jenkins7939498711269983135.bat
C:\ProgramData\Jenkins\.jenkins\workspace\Job-B-Test>echo Running Tests
Running Tests
C:\ProgramData\Jenkins\.jenkins\workspace\Job-B-Test>exit 0
Triggering a new build of Job-C-Deploy
Finished: SUCCESS
Jenkins 2.541.1

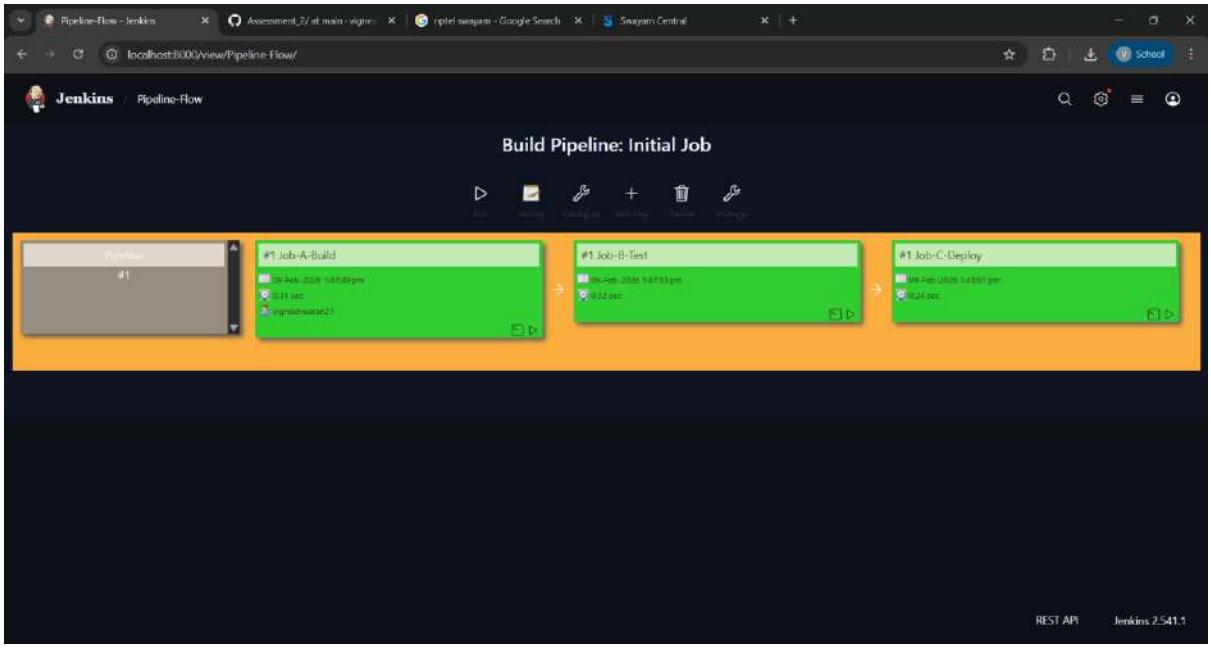

Job-C-Deploy - Jenkins
Assessment_2/ at main · vigne
localhost:8000/job/Job-C-Deploy/
School
Jenkins / Job-C-Deploy
Status
Changes
Workspace
Build Now
Configure
Delete Project
Rename
Builds
Filter
Today
#1 1:48 PM
Job-C-Deploy
Add description
Upstream Projects
Job-B-Test
Permalinks
Last build (#1), 1 min 59 sec ago
Last stable build (#1), 1 min 59 sec ago
Last successful build (#1), 1 min 59 sec ago
Last completed build (#1), 1 min 59 sec ago
REST API
Jenkins 2.541.1



4. Install **Pipeline Stage View**



5. Observe stage-wise execution



Outcome

Learner understands pipeline flow visually.

4. Learning Goal

Use **Jenkins Pipeline & Jenkinsfile**

Stepwise Tasks

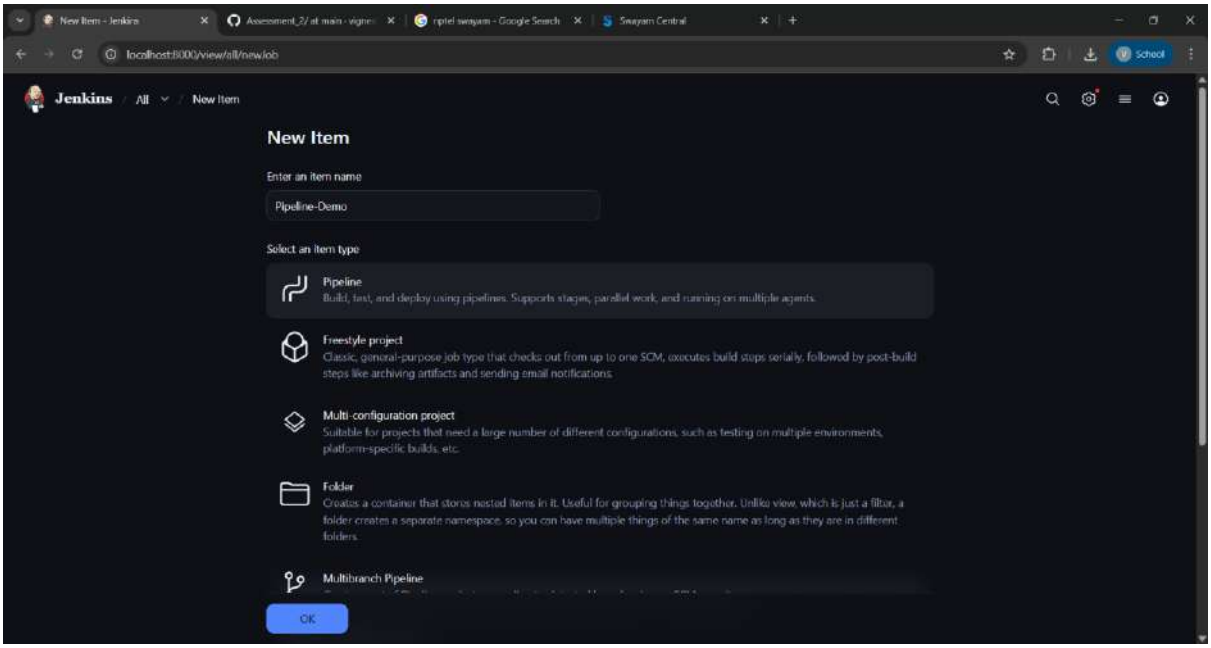
1. What is Jenkins Pipeline?

   A Jenkins Pipeline is **an automated, codified workflow** that tells Jenkins exactly what to do, when to do it, and in what order. It's like writing a cooking recipe for your software delivery process.

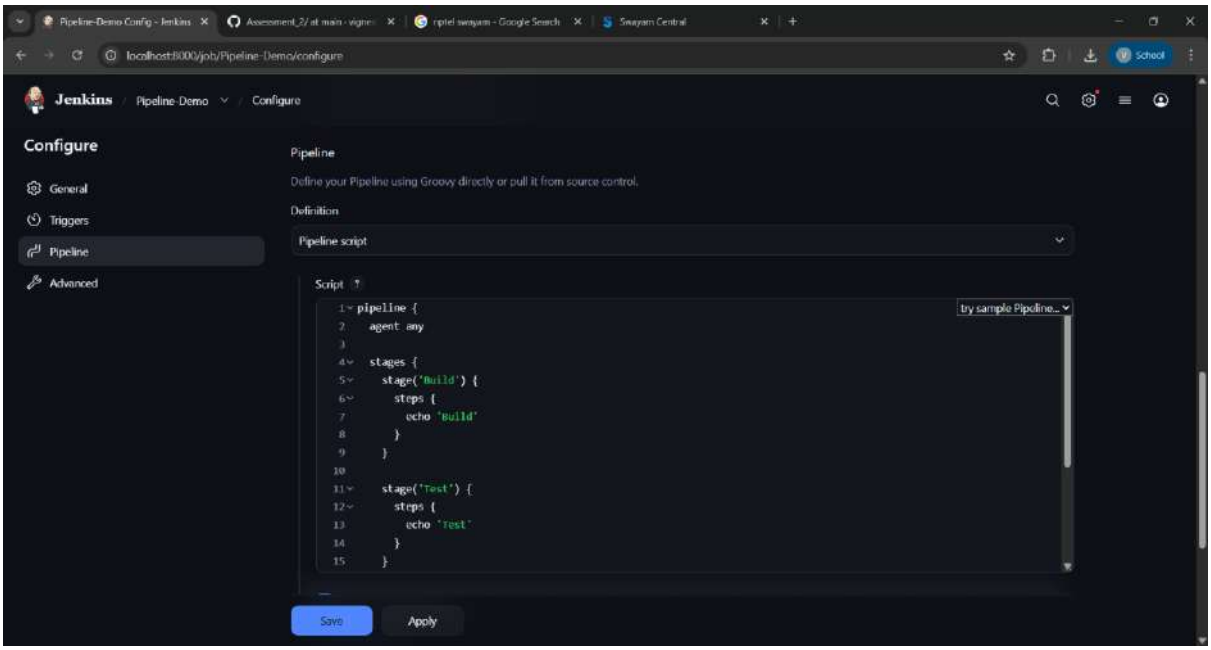
2. Scripted vs Declarative Pipeline

Type	Description	Best For
Scripted Pipeline	Full coding style, very flexible	Advanced users
Declarative Pipeline	Structured, simple, readable	Beginners & most projects

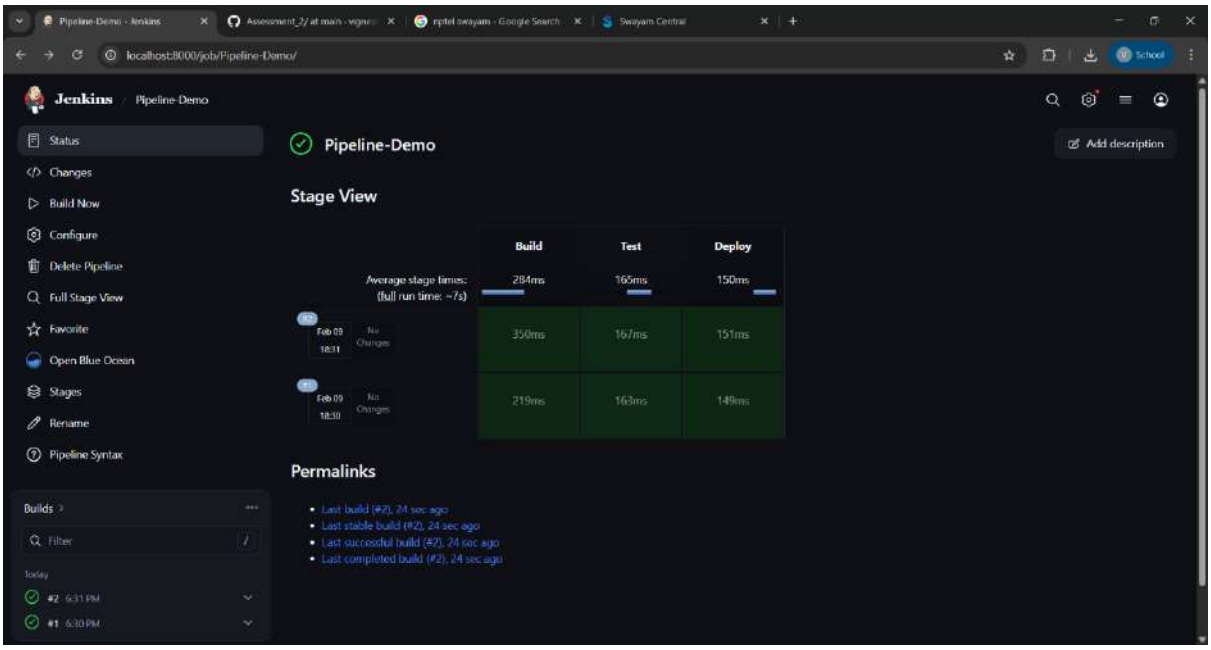
3. Create **Pipeline Job**



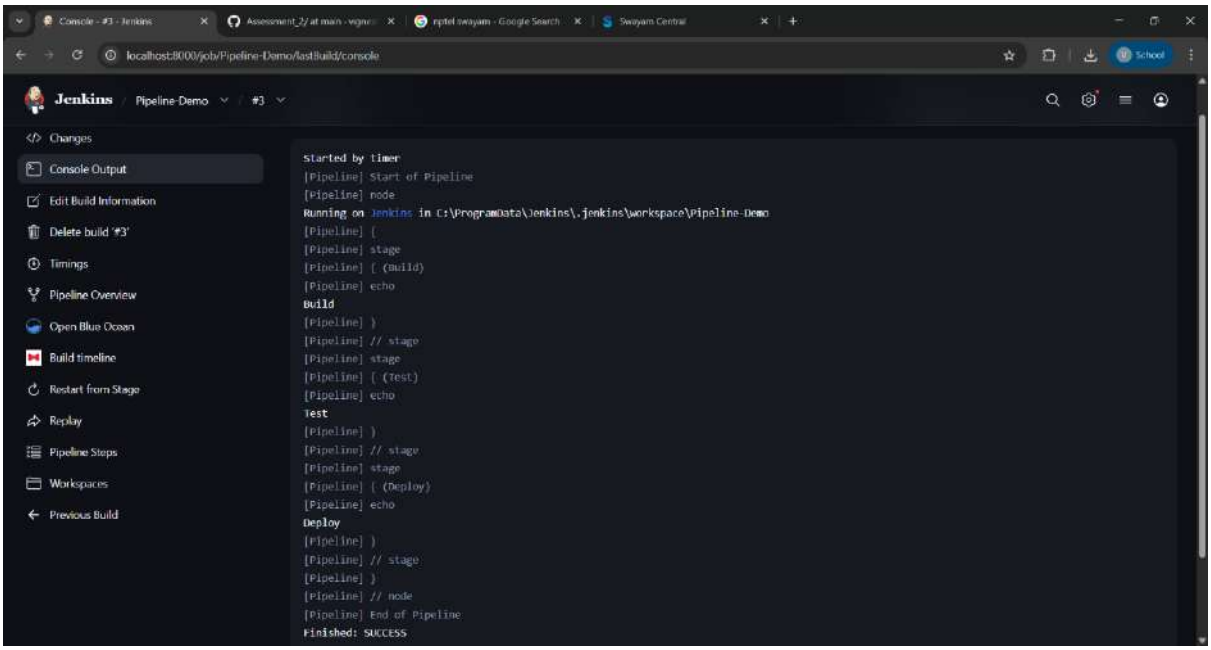
4. Write pipeline in Jenkins UI



5. Add stages:
    - Build
    - Test
    - Deploy



6. Introduce:
    - `agent`
    - `stages`
    - `steps`



Sample Task

```
pipeline {
  agent any
  stages {
    stage('Build') { steps { echo 'Build' } }
    stage('Test')  { steps { echo 'Test' } }
    stage('Deploy'){ steps { echo 'Deploy' } }
  }
}
```

Outcome

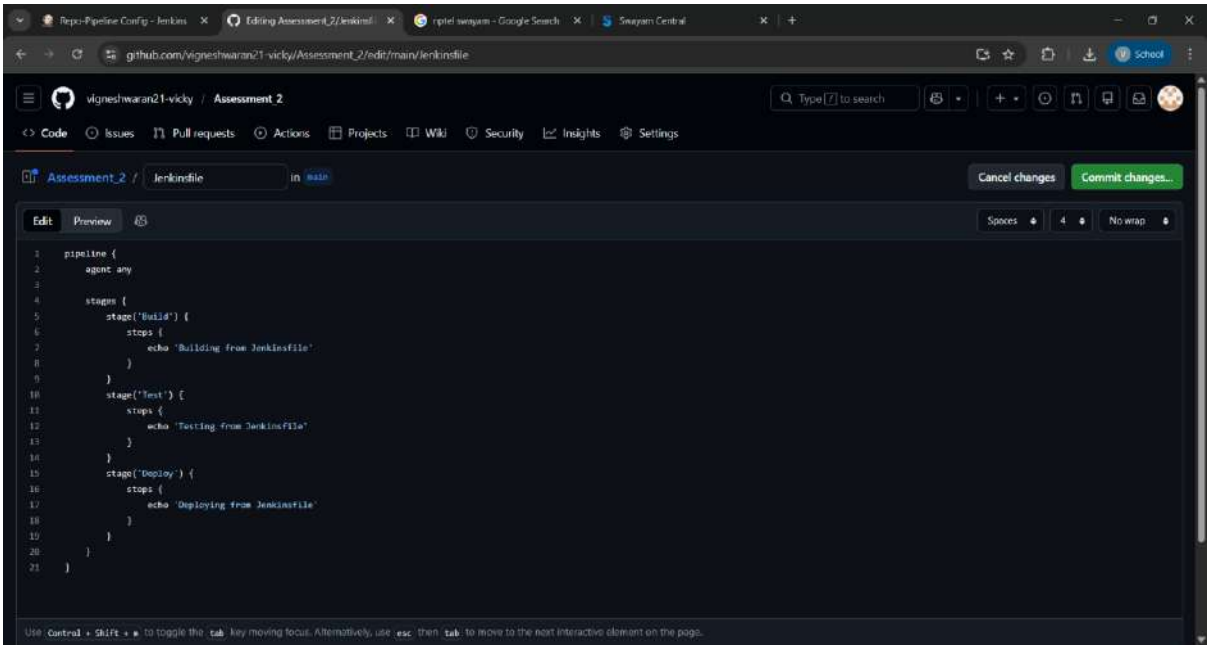
Learner writes basic pipelines confidently.

5. Learning Goal

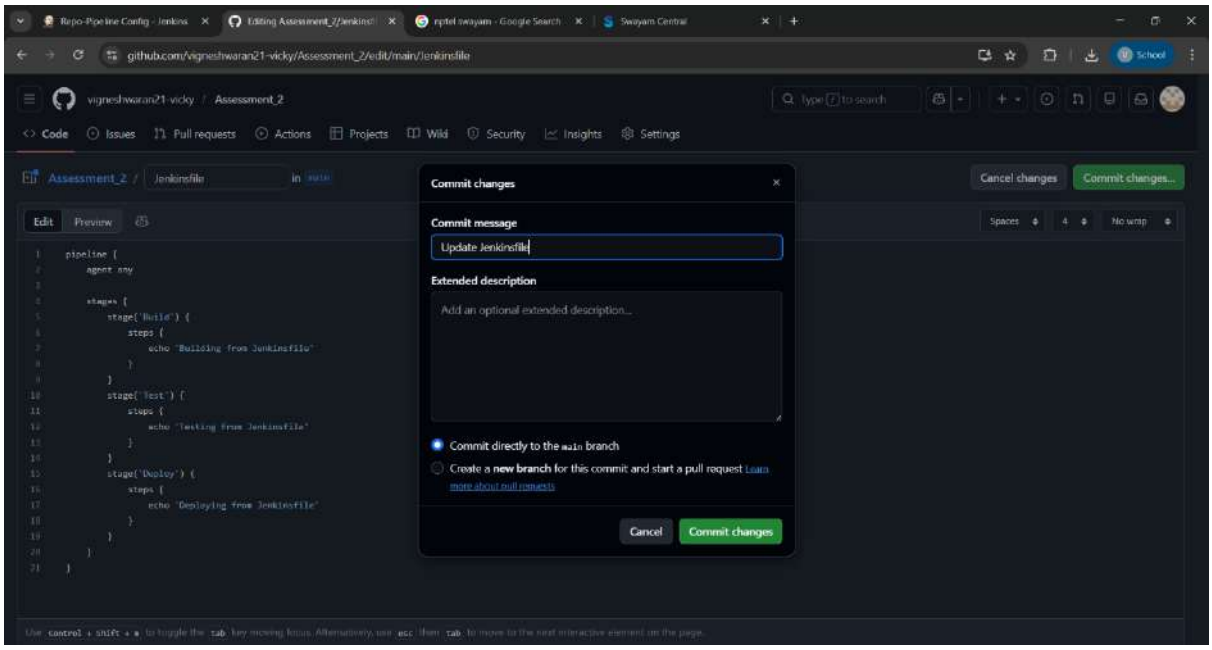
Pipeline as code from repository

Stepwise Tasks

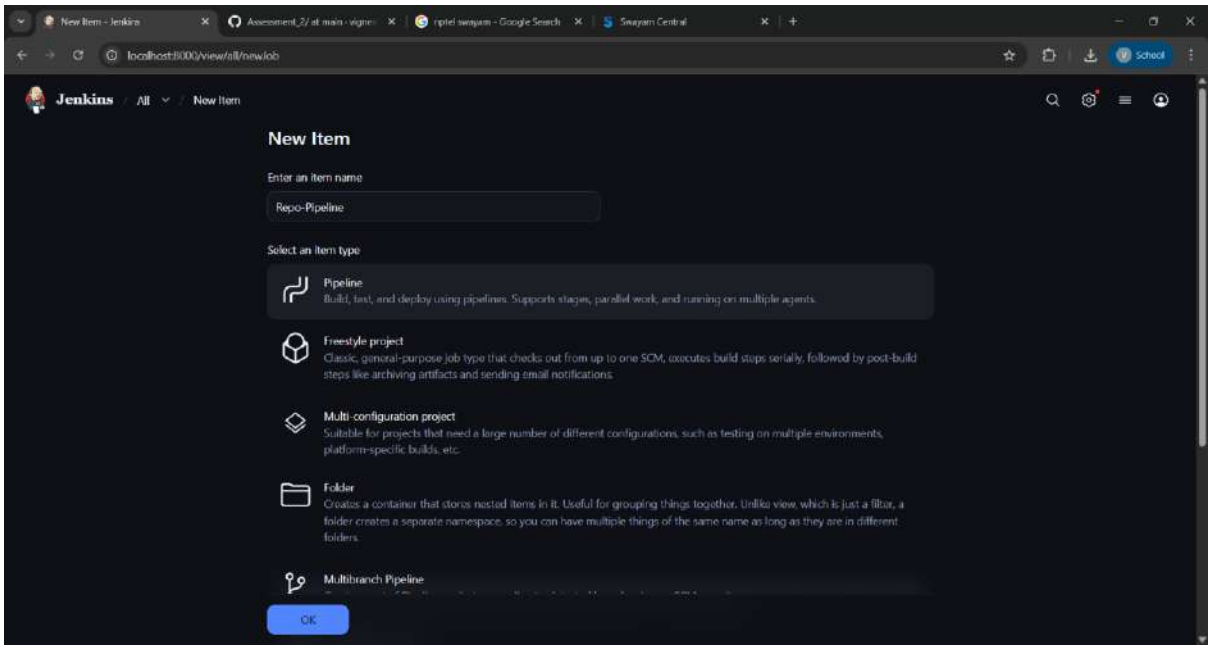
1. Create `Jenkinsfile`

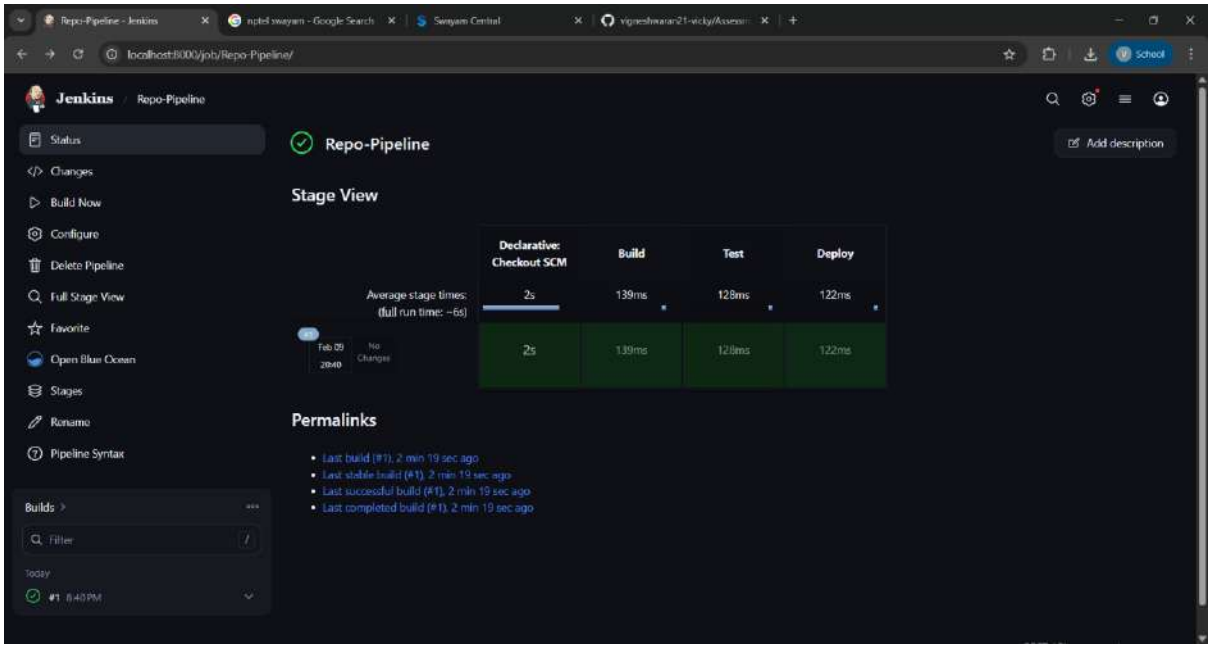


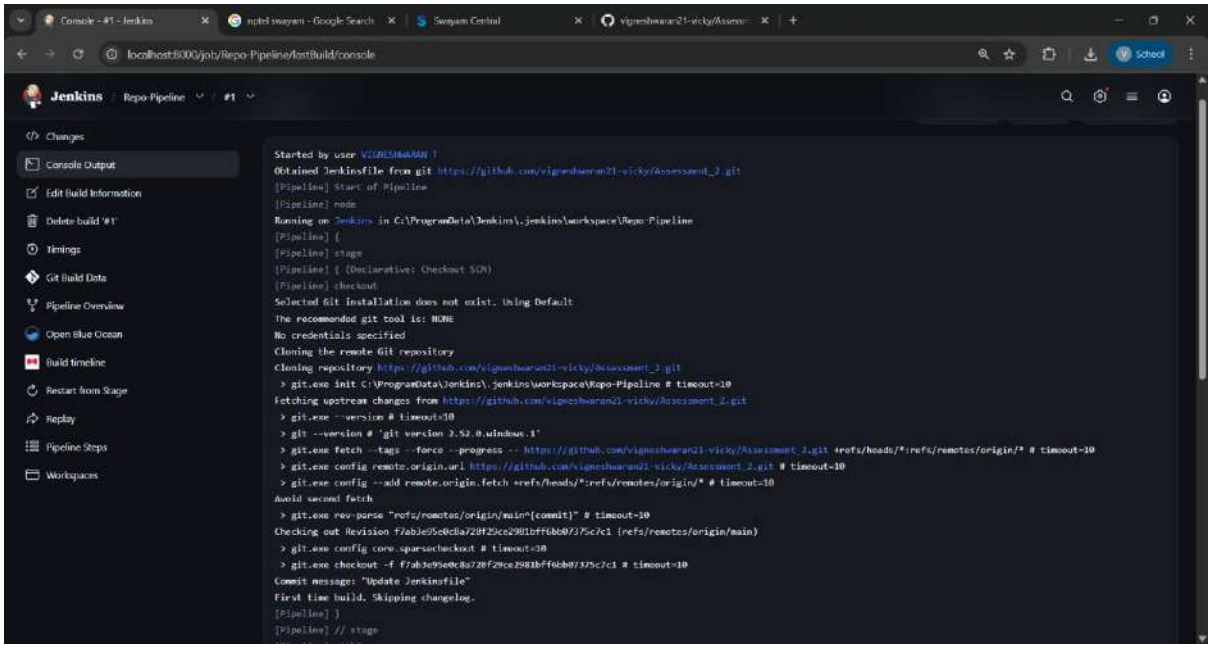
2. Commit Jenkinsfile to GitHub

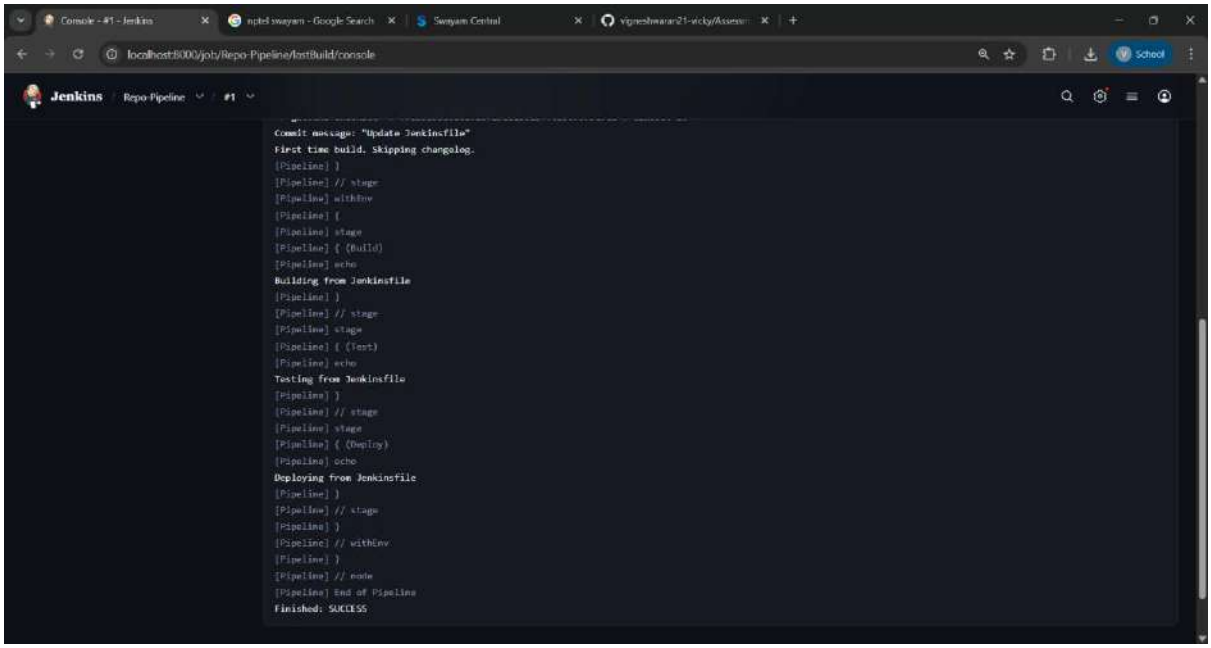


3. Configure Pipeline job:
   - SCM → Git

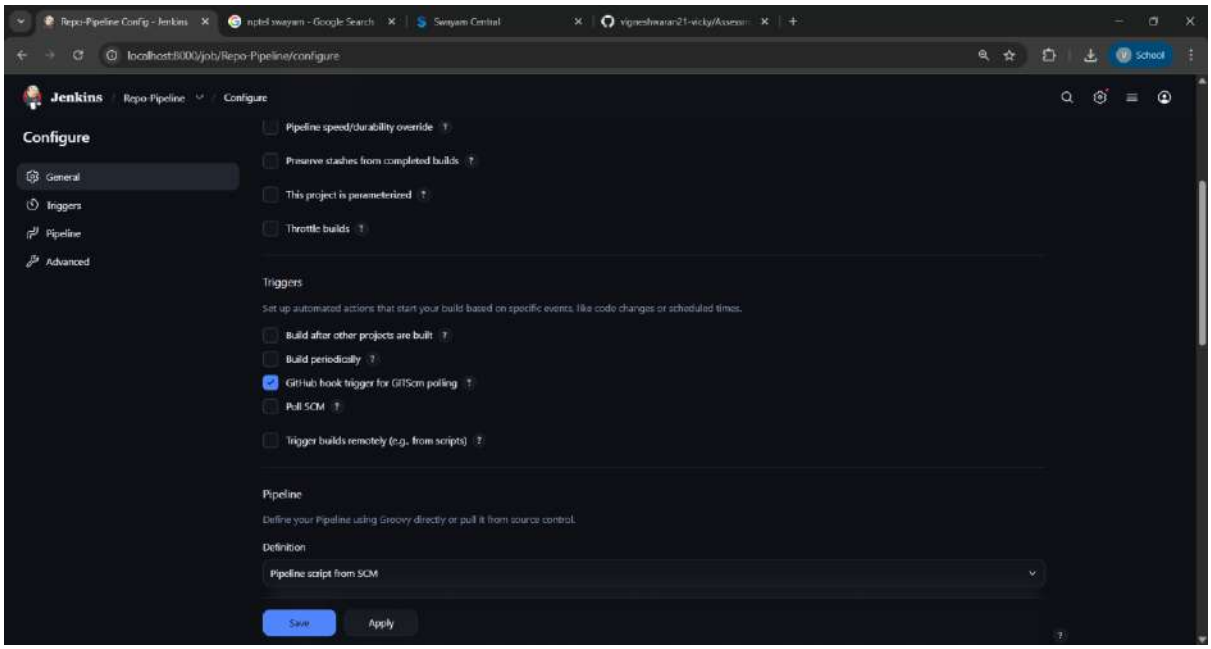



Repo-Pipeline - Jenkins
nptel swayam - Google Search
Swayam Central
localhost:8000/job/Repo-Pipeline/
Jenkins
Repo-Pipeline
Status
Changes
Build Now
Configure
Delete Pipeline
Full Stage View
Favorite
Open Blue Ocean
Stages
Rename
Pipeline Syntax
Builds
Filter
Today
#1 6:40 PM
Repo-Pipeline
Add description
Stage View
Declarative: Checkout SCM
Build
Test
Deploy
Average stage times:
(full run time: ~6s)
2s
139ms
128ms
122ms
#1
Feb 09
20:40
No Changes
2s
139ms
128ms
122ms
Permalinks
Last build (#1), 2 min 19 sec ago
Last stable build (#1), 2 min 19 sec ago
Last successful build (#1), 2 min 19 sec ago
Last completed build (#1), 2 min 19 sec ago

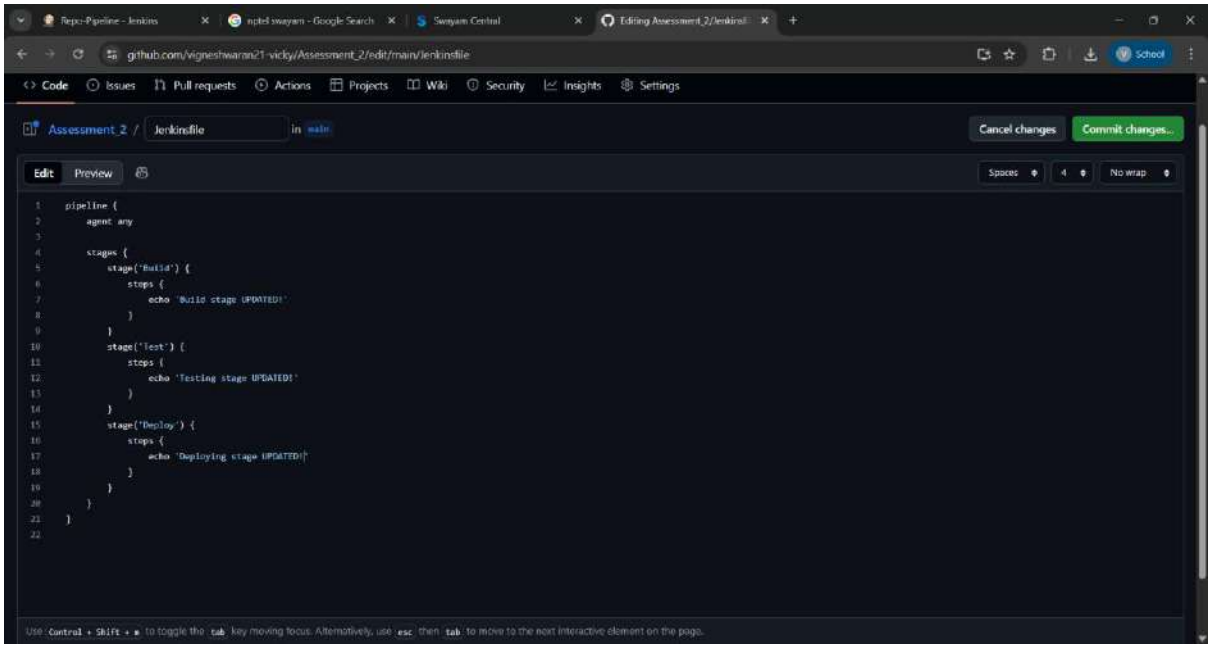

Console - #1 - Jenkins
nptel swayam - Google Search
Swayam Central
localhost:8000/job/Repo-Pipeline/lastBuild/console
Jenkins
Repo-Pipeline
#1
Changes
Console Output
Edit Build Information
Delete build '#1'
Timings
Git Build Data
Pipeline Overview
Open Blue Ocean
Build timeline
Restart from Stage
Replay
Pipeline Steps
Workspaces
Started by user VIGNESHWARAN T
Obtained Jenkinsfile from git https://github.com/vigneshwaran21-vicky/Assessment_2.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\Repo-Pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/vigneshwaran21-vicky/Assessment_2.git
> git.exe init C:\ProgramData\Jenkins\.jenkins\workspace\Repo-Pipeline # timeout=10
Fetching upstream changes from https://github.com/vigneshwaran21-vicky/Assessment_2.git
> git.exe --version # timeout=10
> git --version # 'git version 2.52.0.windows.1'
> git.exe fetch --tags --force --progress -- https://github.com/vigneshwaran21-vicky/Assessment_2.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe config remote.origin.url https://github.com/vigneshwaran21-vicky/Assessment_2.git # timeout=10
> git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision f7ab3e95e0c8a728f29ce2981bff6b607375c7c1 (refs/remotes/origin/main)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f f7ab3e95e0c8a728f29ce2981bff6b607375c7c1 # timeout=10
Commit message: "Update Jenkinsfile"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage

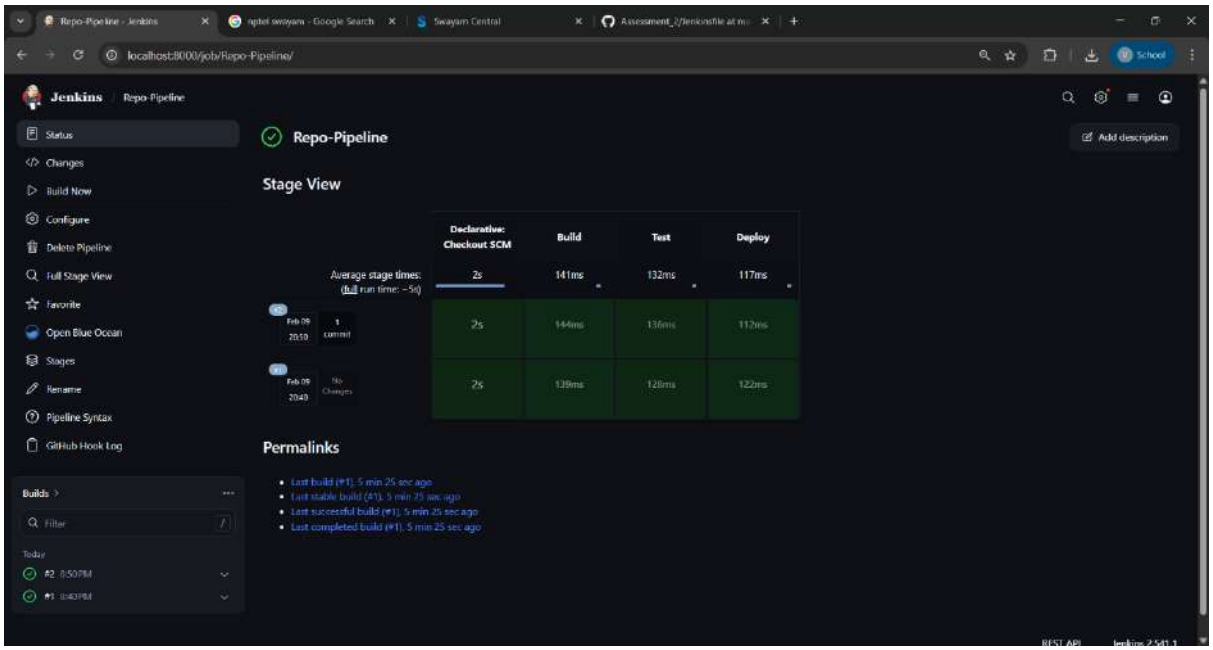


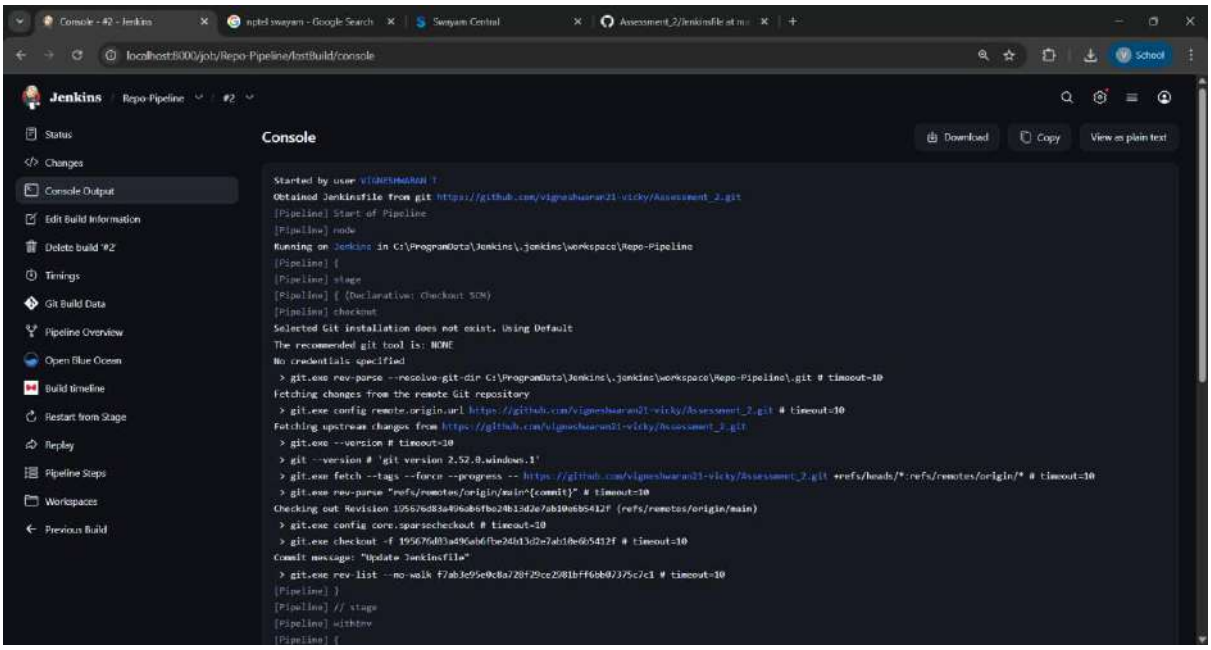
4. Trigger build automatically

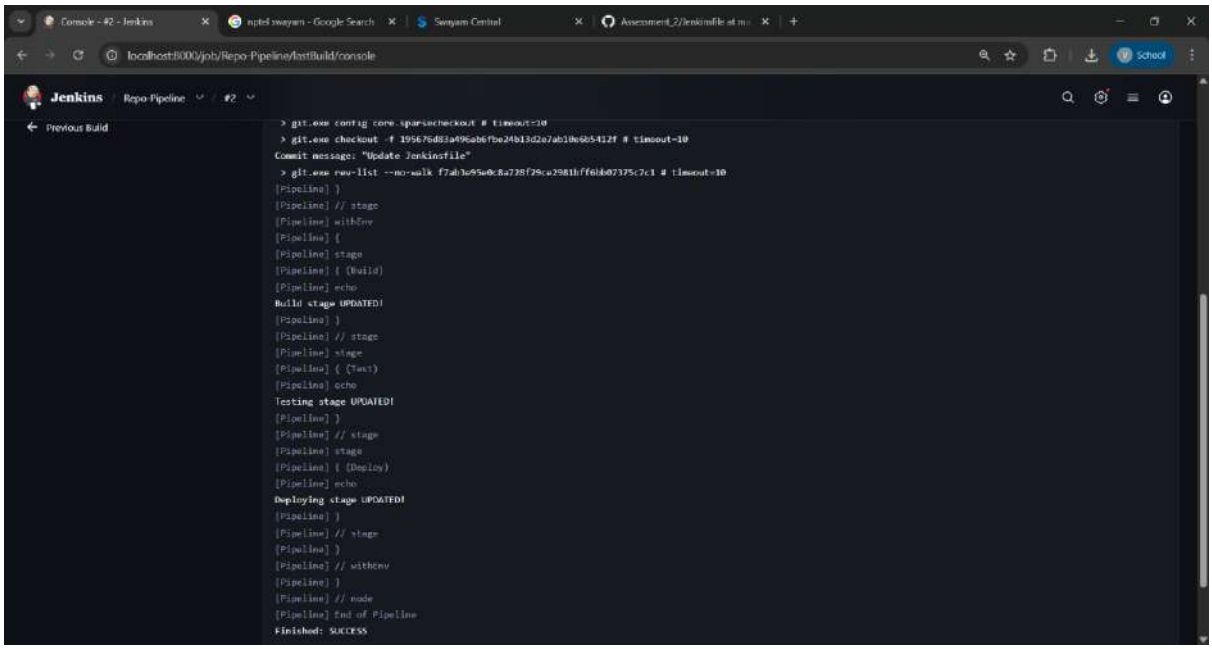


5. Modify Jenkinsfile and rebuild

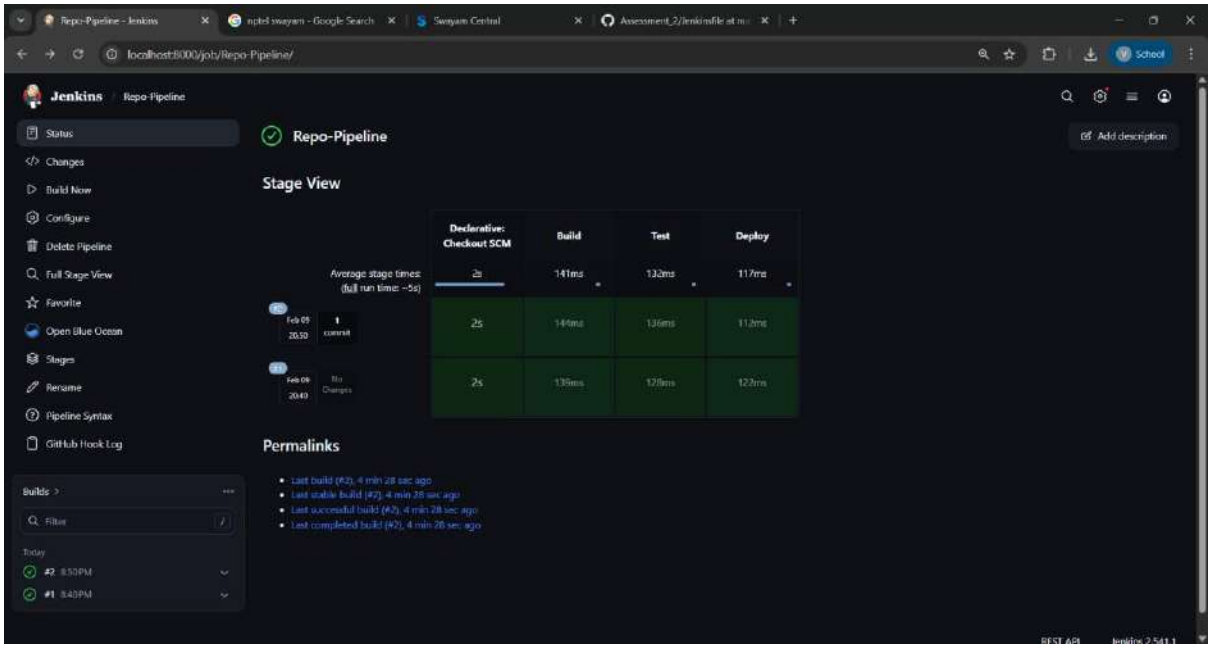





Console - #2 - Jenkins
nptel swayam - Google Search
Swayam Central
Assessment_2/Jenkinsfile at ma
localhost:8000/job/Repo-Pipeline/lastBuild/console
Jenkins
Repo-Pipeline
#2
Status
Changes
Console Output
Edit Build Information
Delete build '#2'
Timings
Git Build Data
Pipeline Overview
Open Blue Ocean
Build timeline
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
Console
Download
Copy
View as plain text
Started by user VIGNESHWARAN T
Obtained Jenkinsfile from git https://github.com/vigneshwaran21-vicky/Assessment_2.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\Repo-Pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
> git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\Repo-Pipeline\.git # timeout=10
Fetching changes from the remote Git repository
> git.exe config remote.origin.url https://github.com/vigneshwaran21-vicky/Assessment_2.git # timeout=10
Fetching upstream changes from https://github.com/vigneshwaran21-vicky/Assessment_2.git
> git.exe --version # timeout=10
> git --version # 'git version 2.52.0.windows.1'
> git.exe fetch --tags --force --progress -- https://github.com/vigneshwaran21-vicky/Assessment_2.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 195676d83a496ab6fbe24b13d2e7ab10e6b5412f (refs/remotes/origin/main)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f 195676d83a496ab6fbe24b13d2e7ab10e6b5412f # timeout=10
Commit message: "Update Jenkinsfile"
> git.exe rev-list --no-walk f7ab3e95e0c8a728f29ce2981bff6bb07375c7c1 # timeout=10
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {


Console - #2 - Jenkins
nptel swayam - Google Search
Swayam Central
Assessment_2/Jenkinsfile at ma
localhost:8000/job/Repo-Pipeline/lastBuild/console
Jenkins
Repo-Pipeline
#2
Previous Build
> git.exe checkout -f 195676d83a496ab6fbe24b13d2e7ab10e6b5412f # timeout=10
Commit message: "Update Jenkinsfile"
> git.exe rev-list --no-walk f7ab3e95e0c8a728f29ce2981bff6bb07375c7c1 # timeout=10
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Build stage UPDATED!
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Testing stage UPDATED!
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying stage UPDATED!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

6. View Stage View



Outcome

Learner practices **Infrastructure as Code** concept.